Hardware Cannibalization and Software Strategy

AI Infrastructure Demand, DRAM Costs, and PC Game System Requirements

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Abstract

The exponential growth of artificial intelligence infrastructure has fundamentally altered semiconductor manufacturing priorities. To meet the bandwidth demands of AI accelerators, memory manufacturers have reallocated fabrication capacity toward High-Bandwidth Memory, constraining the supply of conventional consumer DRAM and interrupting the historical trajectory of declining memory costs. This study examines how this macroeconomic supply shock has affected minimum RAM requirements and optimization strategies in PC game development. Using an Interrupted Time Series design on a sample of 125 AAA cross-platform PC games released between 2015 and 2026, the analysis tests whether the AI demand-driven structural break identified at Q3 2022 altered the trajectory of software bloat and developer optimization behaviour.

The results reveal three key findings: first, a Chow test confirms a statistically significant structural break in PC game RAM requirements at the theoretically predicted point ($p=0.007$). Second, contrary to the predicted suppression of hardware demands, minimum RAM requirements continued to rise after the shock at a rate statistically indistinguishable from the pre-shock trajectory. Third, AI upscaling adoption increased dramatically, with post-shock titles nearly twenty times more likely to support DLSS or FSR than pre-shock counterparts ($p<0.001$).

These findings suggest that when Moore's Law fails at the hardware level, developers layer software-level optimization, such as algorithmic upscaling on top of - rather than in place of - raw memory demands. This refines the theoretical relationship between Wirth's Law and Moore's Law, demonstrating that organizational adaptation occurs through complementary efficiency measures rather than the direct substitution of hardware requirements. Consequently, this study extends Resource Dependence Theory into the empirical context of digital entertainment software development.

Contents

List of Figures

List of Tables

1. Introduction

1.1 Background

The global semiconductor industry is facing one of the most severe shocks in history, leading to the major manufacturers in the supply-side to restructure their production priorities. Historically, the consumer computing market benefited from a consistent trend of declining cost per gigabyte for Random Access Memory (RAM[1]) (Leiserson et al., 2020). This decade long trend was driven by the continuous improvements in manufacturing capabilities linked to Moore's Law (Hennessy & Patterson, 2019). The exponential rise of Generative Artificial Intelligence (GenAI) has fundamentally altered this trajectory.

To meet the huge computational demands of data center Graphics Processing Units (GPUs), semiconductor manufacturers like Samsung, Micron, and SK Hynix have shifted their production toward High Bandwidth Memory (HBM[2]). This is not a different product category; it is an advanced packaging variant of standard Dynamic Random Access Memory (DRAM) technology. HBM is, as a matter of fact, fabricated on the same silicon wafer as the standard consumer memory.

Since HBM and consumer RAM (DDR4 or DDR5) compete for the same limited silicon capacity (TrendForce Research Team, 2026), the rise in AI infrastructure demand has created a "crowding out" effect (H. Kim, 2025). The scaling of HBM has indeed cannibalized the DDR4 and DDR5 supply. This infrastructural and production pivot has constrained the supply capacity for consumer memory, exposing the PC consumer market to significant price volatility and skyrocketing price floors.

The assumption that consumer-created hardware would perpetually improve while declining in cost is no longer a guarantee. This macro shift in semiconductor manufacturing establishes the context for examining how software developers, who had historically strongly relied on hardware abundance, may need to adapt to a world where physical infrastructure constraints and resource scarcity are characterizing the situation.

1.2 Problem Indicator

In the field of Information Systems, digital innovation is governed by a "layered modular architecture" (Yoo et al., 2010), which separates physical hardware components from the digital services and content running on them. This framework has allowed software developer to completely

[1] Refers to the volatile working memory used by systems to store active data (such as game assets) for immediate processing.
[2] Is a premium memory architecture that provides exceptionally high data transfer rates; it is heavily utilized in enterprise AI accelerators.

decouple the code used for programming the software from the underlying hardware constraints. As a result, PC game developers have historically operated under the assumption of cheap, fast-improving hardware that has caused software requirements to get increasingly bloated[3].

The current supply shock of semiconductors has, however, invalidated this assumption. Customers are priced out of necessary hardware upgrades now that the cost of Random Access Memory has risen significantly (TrendForce Research Team, 2026). For digital entertainment firms (specifically game publishers), this creates a barrier to entry, threatening the two-sided network[4] effects that are strictly required for digital platforms to survive. Developers are now facing a serious shrinking of Total Addressable Market (TAM) unless actions are taken. The historic assumption that hardware will eventually get cheaper is no longer a viable foundation for software development strategy (Leiserson et al., 2020; Wirth, 1995).

This research wants to address both a practical and scientific gap. Empirical evidence is needed to understand how developers adapt their game requirements and practices to optimize when hardware cost rises. The study aims, in the end, to provide evidence-based strategic insights for resilient product strategies, in response to recent semiconductor volatility. From an academic perspective, current Information Systems literature focuses on the internal utility of Artificial Intelligence and its adoption inside of organizations. This thesis intends to shift the academic focus from how companies use AI to how they can strategically react and plan for the physical infrastructure constraints and, even more importantly, the resource scarcity caused by it.

1.3 Problem Statement and Research Questions

This research addresses a fundamental problem: there appears to be a lack of empirical evidence regarding how PC game developers adapt their system requirement and optimization practices when the assumption of abundant and cheap hardware falls due to rising prices. To investigate and provide a solution to the problem statement, the following research question is formulated:

Main Research Question: *How has the AI-driven supply shock in the consumer RAM market affected the trajectory of minimum RAM requirements and the adoption of optimization strategies in PC games following the structural break in semiconductor manufacturing priorities?*

And to answer our research question, we divide it into the following sub-questions:

[3] Referring to the tendency of modern applications to become disproportionately resource-intensive, prioritizing rapid feature delivery and code abstraction over memory optimizations.
[4] A market where two distinct user groups provide each other with network benefits (in this case, game developers and consumers).

- **RQ1:** *To what extent has the rise in AI infrastructure demand (e.g., HBM for data-center GPUs) been associated with price anomalies in the consumer RAM market (DDR4/DDR5)?*
- **RQ2:** *How have minimum RAM requirements for high-quality PC games evolved across the pre-AI period versus the AI supply-shock period, and did the trajectory change significantly following the structural break in Q3 2022?*
- **RQ3:** *To what extent have developers complemented raw memory demands with artificial optimization strategies (e.g. AI upscaling technologies such as DLSS/FSR/XeSS) during the AI supply-shock period*[5]*?*

Together, these three sub-questions provide a structured approach to the problem statement. They allow the research to trace the chain of events from AI-driven infrastructure demand and RAM price shocks (RQ1) to changes in PC software requirement trajectories following the structural break (RQ2), and finally to the adoption of artificial optimization strategies (RQ3).

1.4 Research Design Overview

An explanatory, quantitative, and archival research methodology is adopted in order to answer the research questions mentioned above. The research design relies strongly on the triangulation of three quantitative datasets: (i) hardware economic data tracking spot-market indexes for consumer memory, (ii) software specification data extracted via Application Programming Interfaces (APIs) from major gaming databases, and (iii) AI market intelligence data sourced from corporate financial reports.

To isolate the impact of memory price volatility on software requirements, this study employs a quasi-experimental design featuring an Interrupted Time Series (ITS) analysis as the primary empirical strategy. The ITS design models the trajectory of minimum RAM requirements for AAA PC games from 2015 to 2026 and tests whether the structural break at Q3 2022 produced a significant change in the level or slope of requirement growth. This break corresponds to the point at which NVIDIA Data Center revenue overtook Gaming revenue, signalling the large-scale reallocation of semiconductor capacity toward AI infrastructure[6]. This design directly tests whether developers adapted to hardware pricing pressure by suppressing the historical rate of software bloat. To complement this primary analysis, an illustrative Difference-in-Differences (DiD) specification using console counterparts is reported in section 4.6. Rather than a strict causal robustness check, this model

[5] Such as NVIDIA's Deep Learning Super Sampling (DLSS), AMD's Fidelity Super Resolution (FSR) that use ML algorithms to render games at lower resolutions and reconstruct a higher resolution output, reducing raw memory demand.
[6] NVIDIA commands ~80%-90% market share in the AI accelerator segment. Its internal revenues serve as a highly visible and accurate proxy for global AI infrastructural demand shifts.

serves as a structural comparison to demonstrate the asymmetric transmission of the supply shock across open and closed platform governance models.

The study empirically isolates the effect of the supply shock by establishing a baseline trajectory during the extended pre-AI period (2015-2021, with 2022 excluded as a transition year) and comparing it against the AI-intensive period (2023-2026). Furthermore, text mining is applied to historical technical logs to assess the adoption rate of artificial optimization strategies over time.

The complete programmatic pipeline, including the Python scripts used for archival data collection and the R scripts utilized for the econometric estimations, is publicly available for replication purposes at the GitHub link provided in Appendix C.3.

1.5 Relevance

Managerial Relevance

For digital entertainment firms, relying on the assumption of cheap and abundant hardware is no longer a viable product strategy. A central premise of this research is that the rising cost of Random Access Memory creates a barrier to entry for the consumer, which directly threatens the two-sided network effects required for digital platforms to operate successfully (Parker & Van Alstyne, 2005). If users cannot afford the hardware prerequisites, the Total Addressable Market shrinks, subsequently disincentivizing developer participation.

This research provides actionable insights for multiple key stakeholders within and beyond the digital entertainment ecosystem:

- **Game Developers and Publishers:** This study provides a framework for resilient product specification strategies in an era of semiconductor volatility. It assists developers in determining which technical requirements to target to remain aligned with their market's hardware constraints. Furthermore, it informs commercial strategies, such as whether to lower software prices to adjust for the rising component costs faced by consumers, or whether to invest heavily in Artificial Intelligence upscaling to preserve a larger addressable market under volatile memory prices. This directly informs Information Technology business value models (Melville et al., 2004; Brynjolfsson & Hitt, 1996).
- **Consumers (Gamers):** From a societal perspective, this research highlights the growing accessibility barriers in digital entertainment. As hardware costs rise, high-fidelity digital

experiences become progressively more difficult to access, effectively pricing out lower-income demographics and altering the consumer landscape.

- **Gaming Platforms:** For digital distribution platforms, such as Steam, this research illustrates how downstream hardware price shocks threaten the broader platform ecosystem. Understanding these dynamics is crucial for platform owners who must sustain user engagement when the consumer base is unable to upgrade their systems.
- **Policy-Makers:** Finally, the findings hold significant value for regulatory bodies. If the "crowding out" effect caused by Artificial Intelligence infrastructure demand creates a sudden, massive disruption in the broader consumer electronics economy, policymakers may need to consider interventions or actions to mitigate market imbalances and incentivize diversified semiconductor manufacturing.

Theoretical Relevance

Current Information Systems literature focuses heavily on the internal utility of Artificial Intelligence, its organizational adoption, and its value creation within digital platforms (Collins et al., 2021). However, as highlighted by systematic reviews of the field, there is a significant lack of empirical research on the negative economic externalities and physical infrastructural constraints of the Artificial Intelligence boom.

This study addresses this gap by quantifying how the "crowding out" of consumer hardware supply forces a deviation from standard software evolution theories. It shifts the academic focus from how organizations use Artificial Intelligence to how they strategically react to and plan for the resource scarcity caused by it. By linking physical infrastructure constraints (HBM versus DDR4/DDR5 capacity) to software evolution on digital platforms, this study extends Information Systems research on digital innovation and platform governance into the material underpinnings of Artificial Intelligence-driven resource scarcity.

1.6 Structure of the Thesis

To address the central problem statement and, consequently, the research questions, this thesis has five main chapters. Following the first introductory chapter, **chapter two** provides comprehensive review of the literature and the theoretical background. It explores the historic reliance of Moore’s Law, the layered modular architecture of digital platforms, and the concept of software bloat confirmed by Wirth's Law. The synthesis of Resource Dependence Theory (RDT) and the concept of technical debt[7] will close the chapter and help construct the formal research hypothesis.

Chapter three revolves around the quantitative, archival research methodology. It outlines the quasi-experimental design built around an Interrupted Time Series analysis of minimum RAM requirements across the structural break and explains the data collection process via APIs and historical economic datasets. It specifies, furthermore, the exact variables, the primary Interrupted Time Series specification, and the illustrative Difference-in-Differences specification (which compares PC titles against their console counterparts as a control group).

Chapter four presents the results of the analysis. It begins with descriptive statistics and data preparation, followed by a pre-trend visualization and a Chow test validating the structural break. It then reports the formal output of the Interrupted Time Series regression on minimum RAM requirements, a time-series analysis linking Artificial Intelligence market intensity to consumer memory prices, and the adoption trends of AI upscaling technologies. The chapter closes with a Difference-in-Differences illustrative model.

Finally, **chapter five** synthesizes these findings. It provides a non-technical discussion of the results, answers the main research question, and evaluates the theoretical implications for Information Systems literature. The thesis concludes by offering actionable managerial recommendations for product specification strategies, acknowledging the limitations of the study, and suggesting avenues for future research.

[7] Describes the accumulated cost of design shortcuts and unresolved inefficiencies in a codebase. The longer it goes unaddressed, the more costly it becomes to resolve.

2. Literature Review

2.1 Introduction

The theoretical foundation of this thesis is built across five sections, distinct but strongly interconnected. The chapter opens by examining the structural disruption of Moore's Law and the role of Artificial Intelligence infrastructure in displacing consumer memory supply. It then explores how the layered modular architecture of digital platforms made software bloat possible, and how Wirth's Law emerged as the consequence of developers' dependence on hardware improvement cycles.

Subsequently, the chapter examines platform governance dynamics, clearly showing the contrast between volatile PC market and the console environment, the hardware-static one. This is followed by an analysis of how Resource Dependence Theory (RDT), combined with the concept of technical debt explain how rising hardware cost impose a significant organizational pressure on software and game developers. The chapter ends with the synthesis of these theoretical threads into a conceptual model and a set of testable hypotheses that will help guide the analysis of the following chapters.

2.2 The Interruption of Moore's Law and AI Scarcity

For decades the consumer computing market benefited from a stable and enduring economic engine: the steady and continuous decline on cost per unit of computational performance and, most relevant for this study, the cost per gigabyte of Random Access Memory (Leiserson et al., 2020). This trajectory was historically supported by Moore's Law, which projected a doubling of transistor density on integrated circuits around every two years (Hennessy & Patterson, 2019). As transistor density increases, performance improves, markets expand, hardware becomes cheaper, and investment for the new generation of density improvements is financed by the revenue of the current generation (Thompson & Spanuth, 2021). Moore's Law thus operated not merely as a technical projection, but as a self-reinforcing economic cycle. For software developers this cycle translated into an assumption: the hardware for consumers would constantly improve while declining in cost (Wirth, 1995); an assumption that, as this study argues, has been fundamentally disrupted.

Despite this historical trajectory, empirical evidence suggests that the era of the predictable and exponential improvements in processors has effectively ended. Microprocessor performance improvements have sharply declined over the past two decades, falling from an average growth of 52% per year between 1994 and 2003 to a not-so-impressive 4% between 2015 and 2019[8], while

[8] Performance measured using the SPECint benchmark (Thompson & Spanuth, 2021).

performance-per-dollar improvements have plummeted from 48% to 8% over corresponding periods (Thompson & Spanuth, 2021). As transistor scaling yields diminishing returns, the economic cycle described above (markets growing and technical progress mutually helping each other) is showing signs of slowing down (Thompson & Spanuth, 2021). Future performance gains can't, therefore, rely on miniaturization alone but must emerge from higher levels of the computing stack: algorithmic efficiency, software optimization, and specialized architectures (Leiserson et al., 2020). It is exactly this shift towards specialized hardware that has activated the AI-driven supply shock at the center of this study.

This transition towards specialized, domain-specific hardware has seen an aggressive acceleration with the exponential rise of Artificial Intelligence (AI) and Deep Learning (DL) models (Hennessy & Patterson, 2019). Research shows that AI language models predictably get better at their task as a power-law function of model size, dataset size, and the sheer amount of computational power used for training (Kaplan et al., 2020) and because continuously developing AI needs scaling these three factors together, the industry has experienced and insatiable demand for computational power. To meet this demand the industry relies on Domain-Specific Architectures (DSAs) such as GPUs, which process massive matrix multiplications[9] simultaneously within high levels of parallelism (Hennessy & Patterson, 2019). As the processing speed of these accelerators outpaced traditional memory data transfer rates, however, the industry hit a severe performance bottleneck known as "Memory Wall" where fast processors are idly waiting for data to crunch (Gholami et al., 2024). To find a solution to this problem and to feed data to these parallelized accelerators without stalling performance the industry is now using High Bandwidth Memory (HBM) (K. Kim & Park, 2024). By stacking multiple DRAM chips on top of each other using Through-Silicon Vias (TSVs)[10] and advanced packaging, HBM provides an ultra-wide interface able to shorten data transfer paths and time, delivering more than one terabyte per second, which is the massive bandwidth needed to maintain AI accelerators fully utilized (K. Kim & Park, 2024). However, as this study argues, the mass production of HBM has occurred at the expense of conventional consumer DRAM, a consequence that will be examined in the following sections.

As stated, the explosive demand for HBM has generated strong negative externalities for the broader computing market through a direct crowding-out effect. Since HBM and standard consumer DRAM utilize the same silicon wafer capacity, memory manufacturers must now reallocate their limited fabrication lines to prioritize HBM production that offers significantly higher margins than DRAM

[9] Neural network training is computationally dominated by matrix multiplication operations, highly parallelizable.
[10] Vertical electrical connections passing completely through a silicon wafer or die, enabling chip stacking.

manufacturing (TrendForce Research Team, 2026). This upstream market is a very concentrated triopoly dominated by SK Hynix, Samsung and Micron (each accounting for roughly 52.3%, 28.7%, and 19,0% of the HBM market[11]), whose pricing power and focus on AI enterprise clients can sway global supply flows (H. Kim, 2025). According to market analyses, this supply reallocation severely limits the availability of standard consumer and general server memory, sparking a "memory supercycle" characterized by severe capacity shortages and projected DRAM price surges exceeding 70% in 2026 and still growing in 2Q26 (TrendForce Research Team, 2026). These supply constraints are corroborated by measurable price divergences between HBM packages and commodity DRAM, consistent with relationship-specific bargaining dynamics in a concentrated bilateral oligopoly[12] (H. Kim, 2025).

As memory costs surge through the supply chain, the foundational assumption of continuously declining hardware prices is fundamentally broken. Manufacturers of consumer electronics, including PCs and consoles, are taking the hit of the supercycle, having to decide whether to absorb the escalating costs, raise retail prices of their products, or deliberately constrain system memory specs (TrendForce Research Team, 2026). For software and PC game developers, this rising financial barrier threatens to massively shrink the Total Addressable Market (TAM) by pricing out end-users. As consumer hardware costs rise and a portion of end-users are priced out of upgrades, the addressable base of hardware-capable consumers gets smaller and because platform value scales non-linearly with network size, even a modest reduction in the consumer base reduces the incentive for developers to invest in platform-compatible titles in a disproportionate way (Parker & Van Alstyne, 2005). Relying on hardware improvements from the consumers' end to mask inefficient code and "software bloat" is no longer a winning strategy (Wirth, 1995; Xu et al., 2010) and this paradigm shift forces a re-evaluation of system optimization practices and technical debt; a dynamic that will be further explored in the following section.

2.3 Software Bloat and the Hardware Assumption: Wirth's Law, Layered Modular Architecture, and the Moore's Law Dependency

In order to understand why software developers are vulnerable to the hardware supply shock outlined in section 2.2, it's necessary to delve into the logic of modern digital platforms, specifically their architecture. Systems are structured upon a "layered modular architecture[13]" which combines the

[11] Based on TrendForce 2025 estimates as reported in H. Kim (2025).
[12] A market with few sellers and few buyers, where pricing depends on relation-specific negotiations.
[13] A system design principle where software and hardware are organized in distinct horizontal layers, communicating only with adjacent layers through standardized interfaces.

modularity of physical products with the layered nature of digital technology (Yoo et al., 2010). This modular architecture is able to manage complexity by breaking down complex systems into distinct components that interact exclusively through standardized interfaces. While these individual components may still be highly elaborate internally, the architecture hides this complexity behind simple and standardized interaction rules known as interfaces (Baldwin & Clark, 2000). This decomposition allows developers to work on individual modules without having a systematic knowledge of the whole architecture, allowing for specialized division of labour (Baldwin & Clark, 2000). When this modularity is combined with the data homogenization characteristic of digital technologies, it creates distinct layers (devices, networks, services) that follow their own design hierarchies (Yoo et al., 2010). For software developers, finally, this layered modular architecture meant that their code was completely decoupled from any physical hardware constraint (Yoo et al., 2010) and by abstracting[14] the hardware layer, this architectural paradigm made it possible, and even more economically reasonable for developers to prioritize functionality, feature richness and code reusability over computational efficiency (Xu et al., 2010).

The direct observable consequence of this architectural decoupling is Wirth's Law, which states that software grows more resource-intensive at a much faster rate than hardware improves in speed (Wirth, 1995). By operating with a decoupled modular architecture, software vendors feel incentivized to adopt new functionalities without critically thinking about satisfying customers, prioritizing quantity of features over quality of the underlying code. This continuous expansion led to an inevitable "fat software" and a system complexity which is inherently self-inflicted rather than a necessity built to problem-solve (Wirth, 1995). This phenomenon is known as "runtime bloat[15]" (Xu et al., 2010) demonstrating that bloat is native to modern object-oriented applications, based on object-oriented programming (OOP) that relies on deep abstraction and class hierarchies (Wirth, 1995) which encourages developers to prioritize code reusability over performance optimization (Xu et al., 2010). When these abstractions become highly nested and deeply layered, however, they generate massive memory footprints and execution overhead that runtime optimizers can no longer fully mitigate (Xu et al., 2010). Accumulating this software bloat was a reasonable engineering trade-off for developers, who found themselves prioritizing fast development cycles and modular flexibility that created immediate market value, while the inefficiencies of their code were absorbed by the underlying system.

[14] Software developers do not need to understand how physical microchips, memory circuits, or processors operate to write code. The architecture handles the translation between high-level instructions and low-level hardware operations automatically.

[15] The phenomenon where a running program consumes more memory and resources than its core functionality requires

The economic rationality of Wirth's Law was, nevertheless, contingent on the sustained continuity of Moore's Law. The steady decline in the cost of computational performance and, specifically, memory provided the software industry with a "free lunch" for decades (Xu et al., 2010). Software developers "outsourced" performance optimization to the hardware manufacturing supply chain, believing that the next generation of microprocessors and memory would compensate for their increasingly bloated code. This reliance is empirically observable through the phenomenon of "design excess", where the continuous planning of surplus hardware capacity above the immediate software requirements was thought as a design property in order to absorb structural inefficiencies and, ultimately, increase value to the end user (Yadav et al., 2019). As previously stated, this this hardware improvement cycle was so reliable that the software community and developers focussed exclusively on properties like reusability and reliability, correctly assuming that the deficiencies would be handled at the "bottom[16]" of the computing stack. However, recent literature, demonstrates that this age of predictable and constant improvements in processors has effectively ended, therefore forcing future performance improvements to come from the "top[17]" of the computing stack through active software optimization and algorithmic efficiency (Leiserson et al., 2020). As established in section 2.2 the physical limitations of transistor scaling and the supply chain disruption caused by AI specialization have profoundly disrupted the assumption of always-improving, cheaper consumer hardware and when Moore's Law breaks down, the economic and operational rationale of software bloat comes up empty. The external hardware environment can no longer guarantee the necessary design excess to compensate inefficient layered code, thereby exposing the huge "technical debt" that software companies and developers have accumulated throughout decades of relying on deep abstraction and taking shortcuts to get features to market. This structural shift introduces a central theoretical question: does the interruption of Moore's Law (driven by AI hardware scarcity) force a suspension of Wirth's Law?

To explore this theoretical question empirically, this study grounds the abstract dynamics of software bloat and hardware dependence within the specific context of AAA game development. Videogames represent a highly complex intersection between software development and hardware specs, making the gaming development industry an ideal empirical context for measuring the impacts of bloat over time. Modern AAA games are mainly built upon cross-platform game engines[18], which function as extensive software frameworks that are able to abstract the underlying hardware architecture to allow developers to avoid code duplication and focus entirely on creative game design decisions (Politowski

[16] Referring to the foundational physical hardware, specifically the raw microprocessor speed and transistor miniaturization described with Moore's Law
[17] Referring to the software layer including algorithms, high-level programming languages, and application code.
[18] Software frameworks providing essential tools for game development (rendering, physics, audio, and scripting)

et al., 2021). While these engines lower the barrier to entry and facilitate a development that would otherwise be complex and long, they introduce layers of abstraction and generalized interfaces that, following the logic established by Xu et al. (2010), create conditions that may contribute to memory bloat. Since developers utilize these engines to develop code used across many distinct platforms, Random Access Memory (RAM) requirements for PC games represent a reliable, highly visible proxy for the accumulation of software bloat. PC game developers have historically relied on the market's ability to provide continuously hardware design excess to meet the ever-improving memory requirements, an excess that adjusts whenever a static gaming console cycle is released (Yadav et al., 2019) but the recent AI driven price rise in DRAM poses a critical threat to this model, *de facto* shifting the industry from a technology "push[19]" to a technology "pull[20]" (Yadav et al., 2019) and preventing the software industry from continuing to outsource optimization to hardware upgrades. The focus on the next section is on the examination of the different market environment in which PC and consoles operate, necessary to understand how distinct platform governance structures shape development decisions.

2.4 Platform Governance and Market Dynamics

Platforms are not simple neutral technological infrastructures, but they are complex organizations or meta-organizations that federate and coordinate autonomous agents (Gawer, 2014). Federation is essential because bringing together independent innovators into a collective ecosystem cannot be taken for granted. Coordination, on the other hand, is structured through platform governance architectures, which implement control mechanisms and define the boundaries of developer autonomy (Tiwana et al., 2010). Specifically, platform governance addresses the "Goldilocks Governance[21]" problem, which confronts the need to retain sufficient control to ensure the integrity of the platform and the necessity of giving away enough liberty to allow module developers to innovate freely (Tiwana et al., 2010). The behaviour of distinct platforms' developers is strongly influenced by the platform's governance architecture itself, defining their innovation incentives, the way they respond to external dynamics, and technological dependencies (Gawer, 2014; Tiwana et al., 2010). Different governance models - ranging from open ecosystems to closed ones - produce different resource environments and outcomes for the platform they support (Gawer, 2014; Tiwana et al., 2010).

[19] Where excess hardware is pushed onto the market
[20] Where the market must actively demand and pay higher prices for the hardware
[21] Refers to the challenge of fining the right balance, originating from "The Story of the Three Bears" by Robert Southey

The personal computer (PC) market represents an open and volatile platform governance model. That is because by granting access to independent developers and maintaining open technological interfaces, the PC ecosystem creates great innovation incentives and can make use of a broad pool of diverse external capabilities (Boudreau, 2010). This open approach cultivates a vibrant market filled with many different ideas and active experimentation, as the platform governance does not expect to control every piece of the system, but it focuses on opening up complementary development[22] (Gawer, 2014). This openness, nevertheless, delegates a lot of control to the market itself, resulting in a rainbow of consumers hardware configurations that developers and companies must target and somehow accomodate and, since a centralized authority enforcing strict rules is missing, developers operating in this specific ecosystem are directly exposed to upstream macroeconomic supply shocks and external hardware market volatility (Boudreau, 2010; Nieborg & Poell, 2018). The software developer, in this case, has a strong reliance towards the end user ability to navigate and afford the hardware market, resulting in a condition of severe structural vulnerability (Nieborg & Poell, 2018). When physical infrastructure constraints and resource scarcity cause components like RAM to rise sharply in price, the open ecosystem offers no cushion, leaving developers completely vulnerable to fluctuations in consumer DRAM prices.

In contrast, the console market operates under a closed, hardware-static governance model. Console platforms exercise rigid control over the architecture and ecosystem governance, actively managing the interfaces and the overall hardware environment - a governance typology that is the opposite end of the spectrum from the openness of the PC ecosystem (Gawer, 2014). To facilitate coordination between developers and users,, the hardware and memory specification are fixed for the entirety of a console generation[23], providing an environment that is characterized by stable design rules and predictable development conditions, consequently, thanks to this highly regulated ecosystem, software developers are insulated from the fluctuating hardware cost in the component market (Tiwana et al., 2010). This choice has a critical downstream consequence: since the hardware specifications are locked at launch, the platform owner (Sony, Microsoft, Nintendo) assumes full responsibility for managing the entire hardware supply chain and absorbing component price increases, maintaining a stable retail cost for the consumer regardless of the market’s conditions (TrendForce Research Team, 2026). This structural insulation from the memory market renders the console ecosystem a theoretically grounded reference point that clarifies why PC developers are uniquely exposed to consumer DRAM price volatility. While consoles do not serve as the primary

[22] Development of games or services that extend the value of the core platform without being controlled by the platform owner
[23] Usually spanning five to seven years

unit of analysis in this study, they function both as an essential theoretical counterpoint for understanding the asymmetric transmission of macroeconomic hardware shocks across platform governance models and as the control group against which PC titles are compared in the supplementary Difference-in-Differences illustrative structural comparison (section 4.6).

The fundamental difference in platform governance has direct and profound implications for the two-sided network effects that sustain these digital platforms. In a two-sided market, platforms yield true externalities in which the participation of one population directly affects the value and choices of another, linking the value of the platform for developers to the size and active participation of the consumer base (Parker & Van Alstyne, 2005). To overcome the classic "chicken and egg" coordination problem inherent to these markets, pricing structures and subsidies are required to ensure active participation from both sides (Caillaud & Jullien, 2003; Rochet & Tirole, 2003). However, when hardware rise in the open PC market due to macroeconomics events, it acts as a financial barrier to entry, functioning as an anti-subsidy, contracting the consumer base and the TAM for software producers. This shrinking on the consumer side, due to cross-market network externalities, causes the overall value of the platform to decline non-linearly[24], shifting developer incentives to invest in complementary innovation (Boudreau, 2010). While the open PC market transposes the cost shock directly to both developers and consumers, forcing the former to face the consequences of a reducing TAM and reconsider their technical debt (investigated in section 2.5), the console market absorbs the shock entirely. By preserving the TAM through fixed hardware specifications, the closed governance model maintains developer incentives despite macroeconomic volatility.

Understanding how developers strategically respond to this resource pressure requires examining the organizational frameworks that govern their decision-making. This is the focus of the following section.

2.5 Organizational Response: Resource Dependence Theory and Technical Debt

Organizations are not singular entities isolated from macroeconomic tendencies; rather, they are fundamentally integrated with them, and they strongly rely upon external environments to obtain critical resources and their availability (Choi & Phan, 2012; Pfeffer & Salancik, 2015). When the availability of these essential inputs is threatened by supply chain disruption, scarcity or escalating costs, organizational survival necessitates strategic adaptation to either minimize dependency on said

[24] In a traditional one-sided market, losing 10% of a customer base results in a 10% loss in potential sales. In a two-sided market a linear drop in the consumer base causes a compounding (non-linear) collapse in overall value of the platform

resource or secure viable structural substitutes (Pfeffer & Salancik, 2015). When analysing the context of the digital entertainment industry, software developers operate under a profound condition of resource dependence regarding the availability of affordable consumer computing headwear (Choi & Phan, 2012), and the market viability is subordinate to the end-user's ability to navigate the market and acquire the computational power necessary to run selected applications, games, or software. As upstream semiconductor capacity is reallocated towards high margin, Artificial Intelligence data-center infrastructure, the foundational resource of standard consumer Random Access Memory becomes constrained and increasingly expensive (TrendForce Research Team, 2026). Since developers possess no control over upstream DRAM prices or global semiconductor manufacturing priorities, they are forced to strategically adapt to protect their market position (Pfeffer & Salancik, 2015) so, to preserve market access and avoid the collapse of their platform ecosystems, developers are forced to reduce their reliance on raw consumer hardware performance by substituting physical hardware dependencies with internal engineering adaptations (Choi & Phan, 2012; Pfeffer & Salancik, 2015).

For software development organizations, the necessity to adapt to external resource constraints translates directly into a mandate to confront technical debt, representing the compounding consequences of deferred optimization, poorly managed risks, and expedient design choices (Kruchten et al., 2012; Ramasubbu & Kemerer, 2016). For decades, the reliable hardware performance improvements on a regular basis incentivized organizations to tolerate software bloat in order to allow developers to prioritize rapid feature delivery and high-level abstractions over computational efficiency. This continuous trust on the ever-improving hardware layers permitted immense volumes of technical debt to accumulate, embedded within deeply layered, resource intensive architectures that effectively transferred the burden of optimization from the developer's software to the consumer's physical hardware (Ramasubbu & Kemerer, 2016). Very much like financial debt, unoptimized code brings a continuous interest penalty that in this case manifests as quality degradation and deferred investment opportunities that have been masked by the end-user's ability to keep buying faster, larger and cheaper memory modules (Kruchten et al., 2012). The economic rationale for maintaining this debt has now collapsed since the hardware environment has ceased providing the necessary design excess to hide software inefficiencies. The surging cost of consumer RAM creates intense economic pressure that forces organizations to finally address these accrued architectural flaws. Consequently, this pressure translates into an organization command to execute targeted structural maintenance and pay down technical debt through strict codebase optimization (Ramasubbu & Kemerer, 2016). Drawing upon this literature, it is expected that modern game developers will buffer against hardware scarcity by adopting algorithmic efficiency measures, utilizing Artificial Intelligence upscaling technologies as a direct substitute for the raw memory allocations that consumers can no longer afford.

While the macroeconomic pressure to address technical debt is universal across the industry, the capacity of individual organizations to execute this strategy pivot successfully is highly irregular, a difference theoretically explained by the presence and the strength of dynamic capabilities (Teece et al., 1997). Dynamic capabilities define specific organization's strategic routines that are used to integrate, build, and reconfigure internal competencies to address rapidly changing markets and environmental conditions (Eisenhardt & Martin, 2000). A profound reconfiguration of deeply accepted engineering routines is required to allow organizations to transition from a paradigm that is historically tolerant of software bloat to one that focusses on advanced algorithmic optimization (Teece et al., 1997). Organizational capabilities cannot, however, be instantly redirected since massive codebases are built on deep abstraction and pipelines optimized to allow for rapid deliveries represent accumulated asset stocks that are inherently sticky and cannot be loosened quickly and developers face severe time compression diseconomies[25] when attempting to immediately alter their technological capabilities, in stark contrast with the relative ease with which firms can redirect financial investment or reallocate personnel (Dierickx & Cool, 1989). Furthermore, this organization adaptability is heavily constrained by path dependency: historical development choices, foundational architectures, and legacy coding practices create rigid trajectories that strictly dictate how quicky a firm can pivot to optimization-focused practices (Teece et al., 1997). Organizations that lack specific internal routines aimed to rapidly integrate upscaling algorithms or that possess a rigid asset position, will be much slower and will be less effective in responding to hardware scarcity resulting in a highly heterogenous landscape of software adaptation across the market (Eisenhardt & Martin, 2000; Teece et al., 1997).

Synthesizing these theoretical perspectives returns a cohesive framework for understanding how software organizations navigate the current macroeconomic semiconductor supply shock. Resource Dependence Theory establishes the fundamental strategic imperative: as hardware resources become excessively expensive and threaten to shrink the consumer base, developer must actively reduce their structural reliance on brute-force consumer computing power to sustain market viability (Choi & Phan, 2012; Pfeffer & Salancik, 2015) and the specific mechanism for achieving this technological independence requires facing historical accumulations of technical debt (Kruchten et al., 2012). To survive the hardware shock, developers are forced to shift the burden of system performance away

[25] Is a concept from strategic management theory establishing that the accumulation of organizational capabilities cannot be accelerated indefinitely by increasing investment

from the consumer's physical RAM and onto the internal software architecture through codebase optimization and maintenance (Ramasubbu & Kemerer, 2016). Finally, the dynamic capabilities perspective introduces the necessary organizational conditions to achieve this strategic pivot (Eisenhardt & Martin, 2000). The speed and ultimate success of this technical debt reduction will be inherently heterogeneous across the gaming industry, strictly bounded by historical path dependencies, the stickiness of existing asset stocks, and the severe time compression diseconomies that prevent immediate technological transformation (Dierickx & Cool, 1989; Teece et al., 1997). Together, these frameworks establish the theoretical foundation for how the software industry adapts to hardware scarcity. They explain why organizations are economically pressured to confront accumulated technical debt through algorithmic substitution, while also highlighting the severe path-dependent constraints that make this technological transition difficult. Translating these theoretical mechanisms into a testable conceptual model (by examining the suppression of PC system requirements, the temporal adoption of AI upscaling technologies, and their direct correlation with surging consumer DRAM prices) is the focus of the hypotheses developed in the following section.

2.6 Conceptual Model and Hypotheses

Drawing on the theoretical frameworks established in the preceding sections, Figure 1 presents the conceptual model guiding this study. The model traces the causal chain originating from the macroeconomic reallocation of semiconductor capacity toward Artificial Intelligence infrastructure (proxied by NVIDIA data center revenue). This shift creates a mechanism where High-Bandwidth Memory (HBM) production is prioritized, constraining shared fabrication capacity and generating a severe supply-side price shock in the consumer DRAM market. This exogenous resource constraint triggers a definitive structural break (Q3 2022) in the software development environment. Consequently, driven by the imperatives of Resource Dependence Theory, developers are forced to adapt to this shock through two distinct, measurable responses: the suppression of minimum RAM requirements and the substitution of raw physical memory demands with algorithmic efficiency measures, specifically AI upscaling. Finally, the model illustrates that the severity of the consumer DRAM price shock acts as a direct catalyst for this upscaling adoption.

To empirically test the dynamics outlined in the conceptual model, this study proposes four formal hypotheses.

The first establishes an empirical baseline to evaluate the intersection of Resource Dependence Theory and Wirth's Law. It tests whether the structural break caused by the AI-driven supply shock immediately interrupted the historical trajectory of software bloat within the PC ecosystem, or if path-dependent production timelines insulated hardware targeting choices from sudden market volatility.

- **H1 –** Minimum RAM requirements for AAA PC games grew at a significantly lower rate during the AI-intensive period (2023-2026) relative to the pre-AI period (2015-2021), reflecting developer adaptation to consumer DRAM price volatility.

The second hypothesis operationalizes the concept of technical debt, predicting that the required suppression of hardware demands is achieved by integrating specific algorithmic efficiency measures within the affected PC market.

- **H2** – The adoption rate of AI upscaling technologies (DLSS/FSR) among AAA PC game developers increased significantly during the AI-intensive period (2023–2026) relative to the pre-AI period (2015–2021).

While the first two hypotheses evaluate the organizational software response, the underlying theoretical model relies on the existence of the macroeconomic supply shock itself. To empirically validate the origin of this causal chain, the third hypothesis tests the relationship between AI infrastructure expansion and market volatility.

- **H3** – The rise in AI infrastructure demand, proxied by NVIDIA Data Center revenue growth, is positively associated with measurable price anomalies in the consumer DDR4/DDR5 RAM market during the AI intensive period (2023-2026).

Finally, synthesizing these macroeconomic and micro-organizational dynamics, the fourth hypothesis evaluates the direct transmission mechanism predicted by Resource Dependence Theory. It links the severity of the external resource constraint, measured via aggregate production indexes, to the organizational adoption of algorithmic efficiency measures. Testing this relationship allows the study to systematically determine if localized supply chain shocks are observable through industry-wide public metrics, or if they remain masked by broader sector trends.

- **H4** – Higher consumer DRAM prices are positively associated with increased adoption of AI upscaling technologies as a substitution strategy for raw memory demands.

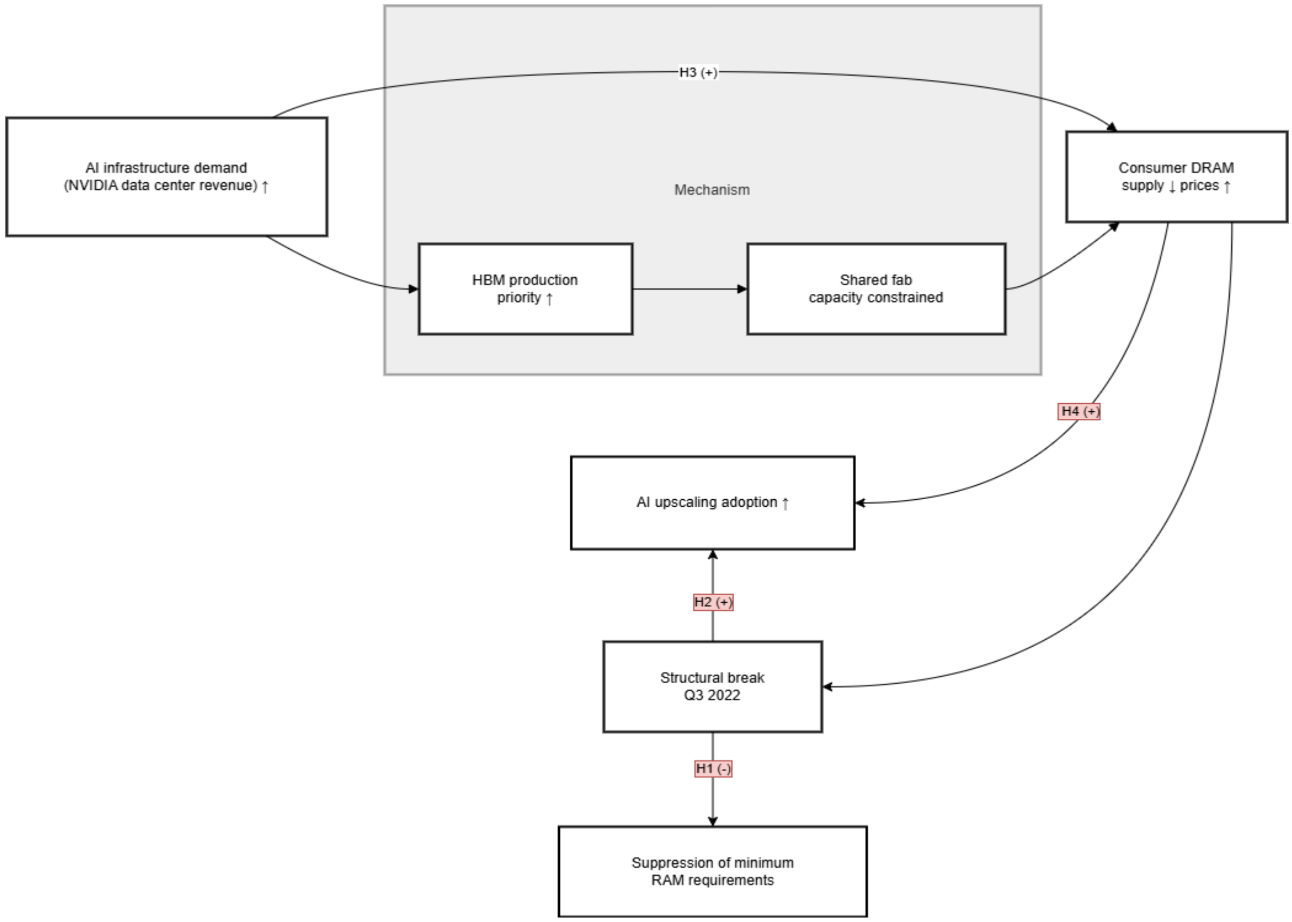

Figure 1 - Conceptual Model

3. Methodology

3.1 Introduction and Research Design

This study employs an explanatory, quantitative, and archival research methodology structured around a quasi-experimental design. The empirical strategy is an Interrupted Time Series (ITS) analysis, which aims to test whether the historical trajectory of minimum RAM requirements for AAA PC games changed significantly following the structural break in semiconductor manufacturing priorities identified in Q3 of 2022. To provide further validation for these findings, a Difference-in-Differences (DiD) analysis (comparing the hardware-fluid PC ecosystem against the hardware-static console market) is utilized as an illustrative structural comparison in section 4.6. This ITS design directly addresses the central theoretical problem established in the literature review: it evaluates whether the rate of software bloat changed when the underlying assumption of abundant, cheap hardware broke down. Specifically, the analysis tests whether the slope of minimum RAM requirement growth changed significantly after the break point. By measuring the slope of hardware demands before and after the AI driven supply shock, the ITS effectively isolates the response of organizations to hardware scarcity. Apart from this primary strategy, the methodology also employs auxiliary analytical approaches to evaluate the causal chain of the conceptual model. Time-series correlations are used to test H3, linking the rise in AI infrastructure demand to consumer DRAM price volatility, and H4, connecting the surging prices to increased adoption of AI upscaling technologies as an organizational substitution strategy.

The remainder of this chapter delves deeper into the execution of this research design. Section 3.2 defines the target sample, outlining the sampling criteria used to isolate AAA titles across the designated pre-AI and AI-intensive periods. Section 3.3 describes the archival data sources and the collection pipeline, detailing the extraction of hardware economic indicators, market intelligence, and software specifications gathered via API from different websites. Finally, section 3.4 specifies the measurement of all the dependent and independent variables, defines the Interrupted Time Series regression specifications, and delineates the analytical procedures used to test each hypothesis.

3.2 Sample Selection

The target population for this study comprises AAA PC games released between 2015 and 2026. This population provides an ideal empirical context for observing organizational responses to hardware scarcity. AAA game development represents the pinnacle of software investment, usually characterized by massive budgets, long development cycles and large engineering teams. Consequently, hardware targeting decisions carry significant strategic weight, as they directly determine the size of the addressable consumer base and the commercial viability of the title. Because modern AAA titles are built upon comprehensive cross-platform game engines that abstract underlying hardware architecture, developers can optimize the software for specific hardware constraints without having to rewrite the core engine code (Politowski et al., 2021). Furthermore, PC game developers gave relied historically on the assumption of hardware design excess to meet escalating memory requirements, an excess that typically adjust whenever a new console generation is released (Yadav et al., 2019). The lower bound of 2015 was selected to provide sufficient temporal depth for reliable ITS slope estimation while capturing the 2016-2018 memory supercycle and avoiding earlier hardware market distortions. The upper bound of 2026 captures the full AI-intensive period following the structural break at Q3 2022. Focusing on this high-investment segment allows the study to effectively observe strategic shifts in software bloat and optimization when the assumption of consumer hardware abundance breaks down.

To construct a rigorous and representative sample, specific inclusion criteria are applied to this target population. First, the definition of a AAA is represented through a publisher-tier approach. The sample includes only title published by industry-leading entities with the financial capacity to execute advanced algorithmic optimization strategies, such as Electronic Arts, Ubisoft, Activision Blizzard, Take-Two Interactive, Square Enix, Sony Interactive Entertainment, Microsoft Gaming, and Capcom. Second, a strict cross-platform requirement is enforced: every PC title that is included in the collection, must possess a direct console counterpart released on the PlayStation 4, PlayStation 5, Xbox One, or Xbox Series X/S. This ensures consistency in production scale and development context across the sample, and preserves the theoretical counterpoint established in section 2.4. Finally, the inclusion time frame is strictly limited from January 2015 to December 2026. Within this window, the third quarter of 2022 (Q3 2022) is designated as the structural break point, separating the pre-AI period (2015-2021) from the AI-intensive period (2023-2026), with 2022 excluded as a transition year.

To preserve the validity of the primary Interrupted Time Series Analysis and the supplementary Difference-in-Differences structural comparison, several exclusion criteria are applied to remove confounding observations. Remasters and remakes of older games are excluded because their hardware requirements often reflect the legacy architecture of the original release rather than contemporary, strategic development decisions. Expansions and downloadable content (DLC) are similarly excluded, as they are not standalone releases and usually inherit the independent system requirements of their base games. “Early Access[26]” titles are, furthermore, omitted from the sample because their system requirements are unstable, frequently unoptimized, and subject to continuous alteration throughout the development cycle. Finally, any games with missing, ambiguous or unverifiable minimum RAM requirements across the selected websites are excluded, as the primary dependent variables of this study require highly reliable and documented quantitative data.

The application of these sampling criteria yields the final dataset utilized for the empirical analysis. The resulting sample comprises 125 AAA cross-platform titles distributed across the 11-year observation window. Of these, 70 belong to the pre-AI period, establishing the baseline trajectory of hardware requirements, while 43 fall within the AI-intensive period, representing the organizational response to the macroeconomic supply shock. 12 observation of 2022, as previously stated, were left out. Table 1 clearly details the exact temporal distribution of the final sample across the observation years. This distribution makes the temporal spread transparent and confirms that the sample is sufficiently balanced to execute the ITS estimation accurately around the structural break.

Year	***Number of titles***	***Period Classification***
2015	10	Pre-AI Period
2016	8	Pre-AI Period
2017	8	Pre-AI Period
2018	7	Pre-AI Period
2019	5	Pre-AI Period
2020	17	Pre-AI Period
2021	15	Pre-AI Period
2022	12	Pre-AI Period (Q1-Q3) / AI-Intensive (Q4)
2023	20	AI-Intensive
2024	14	AI-Intensive
2025	9	AI-Intensive

Table 1 - Sample's game distribution across time

3.3 Data Collection and Sources

The empirical analysis in this study draws on three distinct data streams to construct a comprehensive view of the hardware-software ecosystem. These streams comprise hardware economic data, Artificial Intelligence market intelligence, and software specification data. To ensure reliability and

[26] Refers to a release model wherein a game is made available to the public while still in an unfinished, pre-release testing state.

reproducibility, these distinct data streams are systematically collected and integrated into a unified dataset through an automated data collection pipeline.

The first data stream captured the macroeconomic shifts in the semiconductor market using the Producer Price Index (PPI) for Semiconductor and Related Device Manufacturing. This specific data series (PCU334413334413) is sourced directly from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis (Federal Reserve Bank of St. Louis, 2026). Within the context of this study, the index is utilized as an exploratory baseline macro proxy. The deliberate inclusion of this aggregated series serves an analytical purpose: it tests whether upstream semiconductor manufacturing cost pressures are visible at a macro level, or if localized supply chain shocks are masked by broader sector trends. A recognized limitation is that this index cannot isolate pure consumer DRAM spot prices, as it constitutes a sector-wide average rather than a direct tracking of DDR4 or DDR5 components. However, utilizing the FRED series is academically justified as it is freely accessible, fully reproducible, and establishes a clear public baseline against which the limitations of aggregated economic indicators can be systematically evaluated in the context of AI-driven crowding out effects.

The second data stream leverages corporate financial reporting to quantify Artificial Intelligence market data. The data was manually compiled from NVIDIA's quarterly earnings press releases, covering the period from Q1 2015 to Q3 2025, and structured into a standardized dataset for analysis, segmenting the revenue generated by the Data Center business unit from the Gaming business unit. This financial data captures the AI infrastructure demand proxy. NVIDIA is selected as the specific focus for this metric because the company operates as the dominant manufacturer of AI accelerators, commanding approximately an 80% market share in the data center AI accelerator segment (H. Kim, 2025a). Because of this overwhelming market concentration, NVIDIA's segment revenue functions as a strongly accurate, direct proxy for global AI infrastructure investment intensity. Crucially, the crossover point where NVIDIA's Data Center revenue officially overtook its Gaming revenue in the third quarter of 2022 serves as the exact empirical anchor for the structural break dividing the pre-AI and AI-intensive periods.

The third data stream collects the software specification data required to measure developer responses to hardware scarcity. This information is extracted using the official Steam Store API for each PC title included in the sample (Valve Corporation, 2026). The primary purpose of this extraction is to capture the study's central dependent variable: the minimum RAM requirements, measured in gigabytes (GB), precisely as it was at the time of the game's first release. The data extraction process involves querying specific Steam API endpoints corresponding to game application IDs, with pausing scripts implemented to handle server rate limiting and avoid connection timeouts. A limitation of relying exclusively on the Steam API is that the database can return incomplete, malformed, or missing specification data, especially for titles between 2015 and 2018.

To address the limitations of the Steam Store API, the methodology incorporates targeted manual overrides utilizing PCGamingWiki (PCGamingWiki, 2026), a community-maintained technical database of PC game specifications. This secondary source is used exclusively for titles where the automated Steam API query returns incomplete, missing, or clearly erroneous hardware requirement data. This was a problem particularly prevalent for older titles released from 2015 to 2018. Because the reliability of the dependent variable is paramount, the reliance on PCGamingWiki is heavily regulated. Before any manual override is applied to the dataset, the minimum RAM requirement is manually cross-referenced and verified against at least two independent historical sources.

To enforce the strict sampling criteria requiring architectural parity, the software specification stream is augmented using the Internet Game Database (IGDB) API, accessed via Twitch developer authentication (Internet Game Database, 2026). This secondary API connection captures comprehensive cross-platform release data, which is utilized strictly to verify that each PC title in the sample possesses a qualifying console counterpart. As a technical parameter, the IGDB query pipeline employs a specific category filter set strictly to category=0. Table 2 summarizes the mapping between each hypothesis, the data required to test it, and the corresponding source. This API-level filter ensures that the pipeline identifies only main, standalone game releases, automatically excluding remasters and expansions before they ever enter the local database.

Hypothesis	***Data Used***	***Source***
H1 – RAM Requirements trajectory	Minimum RAM requirements (GB) at initial PC release	Steam Store API, PCGamingWiki
H2 – AI upscaling adoption	Binary DLSS/FSR adoption variable at release	SteamDB technical tag logs
H3 – AI demand and DRAM prices	NVIDIA Data Center quarterly revenue, Semiconductor PPI	NVIDIA Investor Relations, FRED
H4 – DRAM prices and upscaling	Semiconductor PPI, AI upscaling adoption rate	FRED, SteamDB

Table 2 - Hypothesis, data required, and source

The culmination of this data collection methodology is the integration of these three disparate streams into a unified analytical dataset. This process is managed by a custom Python integration pipeline that sequentially fetches, cleans, and stores all collected economic, financial, and software data within a local SQLite database. Once the automated population of the database is complete, the manually verified overrides for missing software specifications are applied via a standardized CSV file before dashboard generation. This final, integrated dataset then serves as the definitive input for the R analysis scripts used to execute the empirical estimations.

3.4 Working Method and Operationalization

The primary dependent variable for this study is the minimum RAM requirement, measured in gigabytes (GB), as specified at the time of each game's initial PC release[27]. This metric is highly valid for evaluating organizational responses to hardware scarcity because it is inherently observable and consistent across titles of different genres and production scales. Most importantly, it directly reflects the deliberate hardware targeting decisions made by development organizations. Because modern cross-platform game engines function as comprehensive frameworks that abstract the underlying hardware architecture (Politowski et al., 2021), the minimum RAM specification operates as a direct, strategic indicator of how heavily a developer relies on hardware design excess to compensate for software bloat (Yadav et al., 2019).

To correctly estimate the empirical models and isolate the effects of the macroeconomic supply shock, the independent and control variables are precisely defined and constructed. The Time Index is a continuous annual variable running from 2015 to 2026, which captures the baseline historical trend in RAM requirement growth. The structural break dummy (Post) is a binary variable coded 0 for the pre-AI period (2015–2021) and 1 for the AI-intensive period (2023–2026). The year 2022 is excluded as a transition year in the primary estimation to avoid within-year contamination from the mid-year structural break identified at Q3 2022, though a sensitivity check including 2022 is reported alongside the main results. Time Since Break is a continuous variable counting the years elapsed since the structural break, coded 0 for all pre-shock years and sequentially as 1, 2, 3, and 4 for the years 2023, 2024, 2025, and 2026[28], respectively. This variable captures the post-shock slope change and serves as the primary coefficient of interest for testing H1. To account for systematic differences in baseline memory demands, genre fixed effects are included as a categorical control variable, adjusting for the fact that certain genres, such as sports titles, consistently require less memory than computationally heavy action-RPG titles. To trace the macroeconomic origins of the shock for H3, NVIDIA Data Center revenue (extracted as quarterly segment revenue in USD millions and aggregated to annual figures) serves as the primary proxy for AI infrastructure demand. The consumer DRAM price proxy

[27] As documented at initial release via Steam Store API or, where necessary, verified through PCGamingWiki manual overrides as described in section 3.3.
[28] With 2026 included conditionally upon the availability of complete annual data at the time of final analysis.

is calculated using the annual average of the FRED semiconductor PPI series, functioning as the core hardware cost proxy to test H3 and H4. This proxy is acknowledged to be imperfect: it captures sector-wide trends rather than isolated DDR4/DDR5 spot prices. Consequently, tests of H3 and H4 are interpreted as exploratory, and the inability to reject or support these hypotheses is attributed to proxy limitations rather than to the absence of a real-world effect. Finally, AI upscaling adoption is defined as a binary variable coded 1 if the title supports DLSS, FSR, or an equivalent technology at release, and 0 otherwise. This serves as the dependent variable for H2 and an independent substitution variable for H4.

It is necessary to contextualize the temporal boundaries of this estimation within the standard production realities of AAA game development. Modern cross-platform titles typically require development cycles spanning two to four years. Consequently, the 2023 to 2026 observation window operates specifically as an immediate transition phase rather than a period of foundational architectural shifts. The primary ITS model, therefore, functions as a direct test of organizational inertia and asset stickiness. It evaluates whether developers possessed the dynamic capabilities to immediately alter their hardware targeting, or if they were locked into memory budgets allocated prior to the macroeconomic structural break at Q3 2022.

To formally test the first hypothesis (H1) and evaluate the trajectory of software bloat, the study employs the following Interrupted Time Series (ITS) regression specification:

$$RAM_{it} = \beta_0 + \beta_1 \cdot Time_t + \beta_2 \cdot Post_t + \beta_3 \cdot TimeSinceBreak_t + \gamma \cdot Genre_i + \varepsilon_{it}$$

In this equation RAM_{it} represents the minimum RAM requirement for game i in year t. The coefficient β_0 serves as the intercept, establishing the estimated baseline RAM requirement at the start of the observation window. The coefficient β_1 captures the pre-shock slope, quantifying the historical annual rate of RAM requirement growth prior to the structural break. The coefficient β_2 isolates the level change at the break point, measuring whether RAM requirements experienced an immediate jump or drop exactly at Q3 2022. Crucially, the coefficient β_3 captures the post-shock slope change and functions as the primary coefficient of interest for testing H1. A negative and statistically significant β_3 indicates that the rate of RAM requirement growth slowed following the macroeconomic supply shock, a result consistent with developer adaptation to hardware scarcity. Furthermore, γ captures the genre fixed effects to account for unobserved, time-invariant differences in memory demands across different game categories, while ε_{it} represents the stochastic error term.

Based on this empirical specification, the formal statistical tests for the first hypothesis are the following:

- **H1 Null Hypothesis ($H1_0$):** $\beta_3 = 0$. There is no significant change in the slope of RAM requirement growth after the structural break.
- **H1 Alternative Hypothesis ($H1_A$):** $\beta_3 < 0$. The slope of RAM requirement growth declined significantly after the structural break.

A one-sided alternative is theoretically motivated. Resource Dependence Theory and the technical-debt mechanism jointly predict that developers respond to a binding hardware-cost constraint by suppressing the rate of software bloat. That counts a directional prediction, not a symmetric one. Within this framework, β3 ≥ 0 is not a region of the parameter space that the conceptual model predicts. Observation of β3 > 0 therefore rejects H1A by sign, independently of the magnitude of the standard error: such an outcome would be evidence against the directional adaptation hypothesis rather than insufficient evidence for it.

The model is estimated using Ordinary Least Squares with heteroskedasticity-robust standard errors, implemented in R using the *fixest* package. The significance threshold is set at $p < 0.05$ throughout the analysis.

The analytical procedure for testing the hypotheses proceeds in five sequential steps. First, a pre-trend visualization is conducted by plotting the annual mean minimum RAM requirements across the entire observation window. This graphical analysis serves to visually confirm the baseline trajectory of hardware demands and identify any obvious discontinuities surrounding the supply shock. Second, a Chow test is conducted to confirm the third quarter of 2022 (Q3 2022) as the true structural break point. If the Chow test confirms a statistically significant break at this moment, the analysis proceeds using this theoretical break date, and both the test statistic and p-value are reported. Third, the main Interrupted Time Series regression is executed. The segmented regression specification defined above is estimated using the *fixest* package in the R programming environment, applying heteroskedasticity-robust standard errors. The results report the estimated coefficients for the pre-shock slope (β_1), the level change (β_2), and the primary post-shock slope change (β_3), alongside their respective standard errors and p-values. Fourth, a genre subsample analysis is conducted to assess whether the organizational response is heterogeneous across different software categories. The ITS regression is run separately for computationally lighter sports titles and graphically intensive titles, such as action-RPGs, to observe potential variations in adaptation strategies. Fifth, a sensitivity check to evaluate the robustness of the findings is performed. This involves re-estimating the model by incorporating the excluded transition year of 2022 as either entirely pre-shock or entirely post-shock and comparing these alternative results against the main specification.

To test the second hypothesis (H2) regarding the organizational response to technical debt, the analysis employs a logistic regression model[29]. AI upscaling adoption serves as the binary dependent variable, evaluated against the structural break period dummy as the primary independent variable. To account for systematic differences in baseline graphical intensity and computational demands, the model includes game genre as a categorical control variable. The results report the estimated odds ratio and the statistical significance of the period coefficient to determine whether the likelihood of integrating upscaling technologies increased significantly during the AI-intensive period.

Finally, to evaluate the macroeconomic and substitution mechanisms outlined in the third and fourth hypotheses (H3 and H4), the study conducts time-series correlation analyses. For H3, the analysis tests the relationship between AI infrastructure demand and hardware costs by calculating the Pearson correlation coefficient between annual NVIDIA Data Center revenue and the annual FRED semiconductor PPI values, reporting its statistical significance[30]. Where distributional assumptions are violated, Spearman rank correlation is reported as a non-parametric alternative. To estimate the exact magnitude of this association, a simple Ordinary Least Squares (OLS) regression of the PPI on NVIDIA revenue is conducted. For H4, the identical analytical approach (reporting the Pearson correlation coefficient, significance, and a simple OLS regression) is applied to test the relationship between the annual FRED PPI values and the annual AI upscaling adoption rates.

As an illustrative extension of the primary findings, a Difference-in-Differences specification is reported in section 4.6. It is necessary to explicitly acknowledge that this model does not satisfy the strict parallel trends assumption required for causal identification. Because console observations are coded at a fixed 16GB capacity dictated by static hardware design, they cannot organically respond to DRAM price volatility. Therefore, this specification is not utilized as a strict causal robustness check, but rather as a descriptive measure of structural divergence. It quantifies exactly how much the open PC ecosystem deviated from the insulated baseline of the console ecosystem following the macroeconomic supply shock. The formal DiD regression equation is specified as follows:

$$RAM_{it} = \alpha + \beta_1 \cdot PC_i + \beta_2 \cdot Post_t + \beta_3 \cdot (PC \cdot Post)_{it} + \gamma \cdot Genre_i + \varepsilon_{it}$$

In this specification, β_3 represents the primary DiD coefficient of interest, capturing the differential change in PC RAM requirements relative to the insulated console requirements following the macroeconomic supply shock.

[29] The model is estimated in R using the *glm* function with a binomial family specification.
[30] A significance threshold of $p<0.05$ is applied.

4. Results

4.1 Descriptive Statistics and Data Preparation

The final dataset for this study comprises 125 observations spanning an 11-year observation window from 2015 to 2026, with the year 2022's games explicitly excluded from the primary estimation. The structural break identified by the Chow test occurs at Q3 2022, meaning the shock falls mid-year. Including 2022 would require classifying it as either entirely pre-shock or entirely post-shock, both of which would misclassify approximately half of the year's observations and introduce artificial contamination into either the baseline trajectory or the post-shock period. Excluding it produces a cleaner separation between the two periods. A sensitivity check re-estimating the model with 2022 included as a pre-shock year is reported in section 4.3 and confirms the findings are robust to this boundary decision. The sample is divided into 70 pre-shock observations drawn from the 2015–2021 period, and 43 post-shock observations drawn from the 2023–2026 period. The titles in the dataset are distributed across four major gaming genres: Action-Adventure, Action-RPG, First-Person Shooter (FPS) and Sports. It should be noted that there is an uneven temporal distribution within the sample. There are significantly more observations in the pre-shock period by design, as the historical baseline spans seven years while the available post-shock observation window is much shorter.

Descriptive statistics reveal a clear shift in hardware demands across the two observation periods. Overall, the mean minimum RAM requirement for the sample shifted from 7.06 GB during the pre-shock period to 11.60 GB in the post-shock period. This represents a raw increase of 4.54 GB, approximately 64.3% across the two periods. Furthermore, a breakdown of the data by genre highlights significant heterogeneity in hardware demands across different software categories. Most notably, Action-RPG titles exhibited the highest hardware demands, reaching a mean minimum RAM requirement of 13.1 GB in the post-shock period. Conversely, Sports titles maintained the lowest hardware footprint, averaging a minimum requirement of just 8.2 GB post-shock. Table 3 provides the complete breakdown of mean RAM requirements categorized by both genre and period.

Genre	*Pre-AI*	*AI-Intensive*	*Change*
Action-RPG	8.00	13.10	+5.10
Action-Adventure	7.08	12.30	+5.22
FPS	7.88	12.00	+4.12
Sports	6.09	9.00	+2.11
Overall	**7.06**	**11.60**	**+4.54**

Table 3 - Mean RAM Requirements by Genre and Period (GB)

4.2 Pre-Trend Visualization and Structural Break Validation

To establish the historical context of hardware demands, Figure 2 illustrates the long-run trajectory of minimum RAM requirements across major PC game franchises from 2002 to 2022. The chart demonstrates a consistent upward growth in memory demands over two decades, firmly aligning with the theoretical predictions of Wirth's Law. Notably, this historical bloat culminates in a clear convergence around the 8.0 GB mark across multiple major franchises between 2019 and 2022, establishing a distinct pre-shock baseline. Observing this steady, long-run trend provides the theoretical and empirical foundation for expecting a stable pre-shock slope in the subsequent Interrupted Time Series estimation.

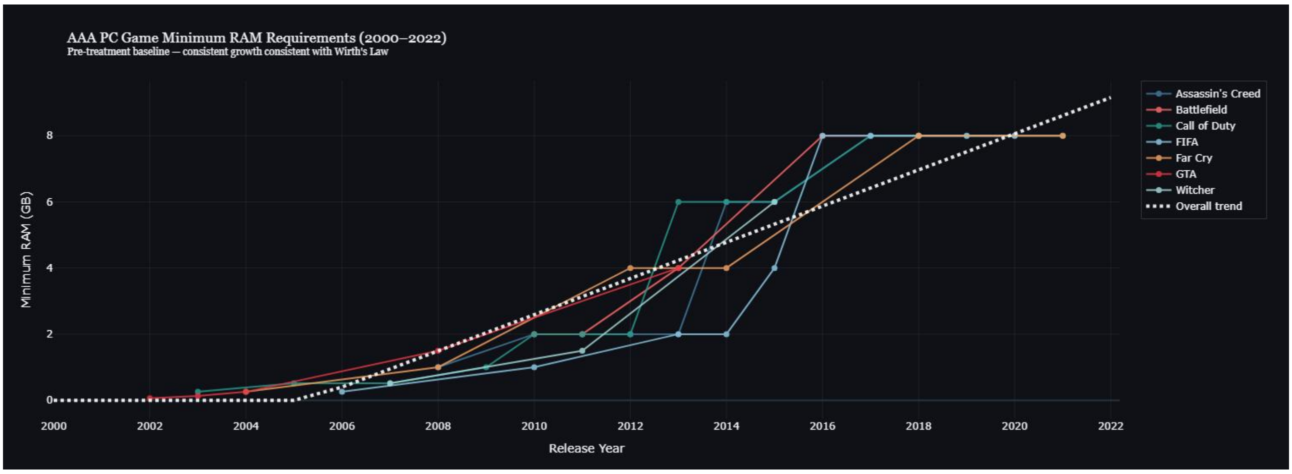

Figure 2 - pre-trend trajectory of RAM requirements (2002-2022)

Building upon this historical baseline, Figure 3, presents the actual ITS dataset visualization spanning the observation window from 2015 to 2026. The plotted annual means reveal a clear trajectory: from 2015 to 2021, minimum RAM requirements exhibited relatively stable growth, moving from an average of 6.0 GB to 7.0 GB with minor fluctuation. This translates to a mean annual growth of approximately 0.18 GB per year, which is consistent with the pre-shock slope (β_1) estimated in the ITS model. In contrast, the period from 2023 to 2025 shows a sharp acceleration in hardware demands. Mean requirements jumped significantly, rising from 9.56 GB in 2023 to 13.33 GB across 2024 and 2025. This severe visual discontinuity aligns perfectly with the structural break hypothesized in the conceptual model. As specified in the methodology, data points for the year 2022 are deliberately excluded from this primary plot to prevent transition-year contamination around the exact mid-year shock.

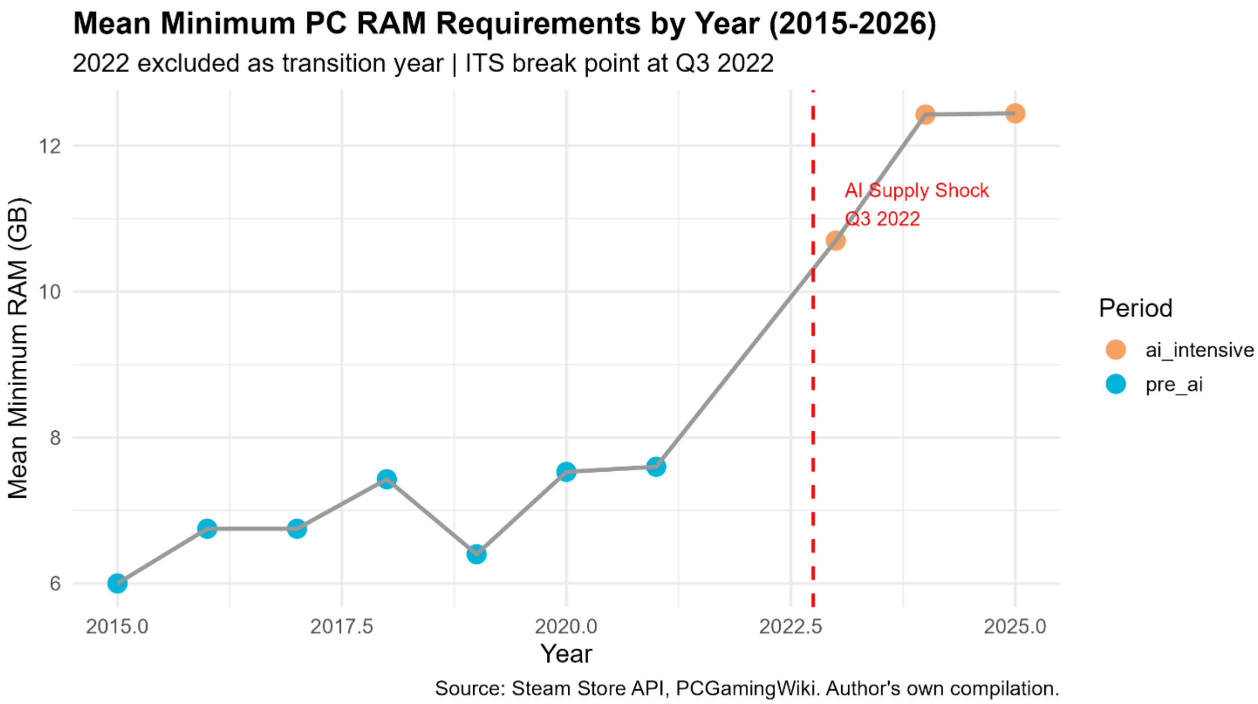

Figure 3 – Mean Min. PC RAM Requirements by year

To formally validate this visual discontinuity, a Chow test was conducted to evaluate the presence of a structural break at the designated point. The test yielded highly significant results ($F = 5.206$, $df = 2$ and 109, $p = 0.007$). This indicates that the unrestricted model, which allows for different slopes and intercepts before and after Q3 2022, fits the data significantly better than a restricted single-trend model. This provides strong statistical confirmation that a structural break in PC game RAM requirements occurred exactly at the theoretically predicted point. The Chow test result empirically validates the Q3 2022 structural break date identified from NVIDIA segment revenue data, confirming that the periodization adopted in this study is justified. The structural break is visually confirmed in Figure 4, which illustrates the divergence between the stable pre-shock RAM growth trajectory and the accelerated requirements observed in the AI-intensive period.

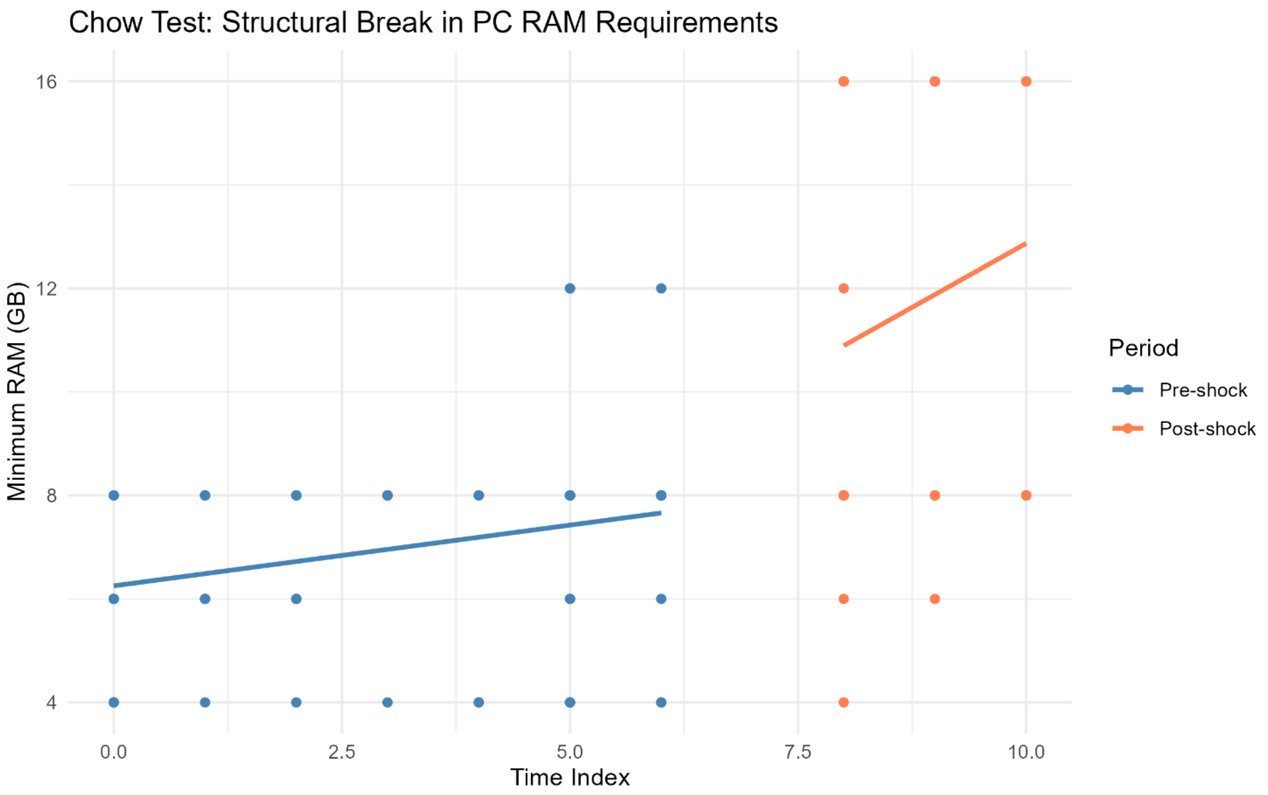

Figure 4 - Chow Test Validation of structural break at Q3 2022

Together, the pre-trend visualization and the Chow test provide convergent evidence that PC game minimum RAM requirements underwent a statistically significant structural change at the theoretically predicted break point. This finding justifies the application of the ITS framework and establishes the baseline trajectory against which post-shock developments are measured.

4.3 Interrupted Time Series Analysis: RAM Requirement Trajectory (H1)

To evaluate the structural break in hardware demands, three distinct Interrupted Time Series (ITS) model specifications were estimated. Model 1 presents the basic ITS regression with no additional controls. Model 2 incorporates genre fixed effects to account for baseline differences in computational and graphical intensity across game categories. Model 3 further adds recommended RAM as an additional scope control to adjust for the overall scale and fidelity of each title. Among these, Model 2 serves as the preferred specification because it successfully controls for systematic genre differences while maintaining the complete sample of 113 observations. In contrast, Model 3 loses 7 observations due to missing recommended RAM data within the historical archives. Table 4 presents the coefficient estimates and Standard Errors for all three models side by side.

	Model 1 – Basic ITS	***Model 2 – ITS with Genre FE***	***Model 3 – ITS with Genre FE + Rec RAM***
Pre-shock slope (β_1)	0.235 (*0.097*) *	0.268 (*0.091*) **	0.084 (*0.107*)
Level Change (β_2)	2.007 (*1.620*)	1.596 (*1.437*)	1.766 (*1.334*)
Post-shock slope change (β_3)	0.755 (*0.813*)	0.637 (*0.714*)	0.357 (*0.674*)
Intercept (β_0)	6.253 (*0.348*) ***	FE	FE
Adjusted R^2	0.369	0.457	0.533
Genre Fixed Effect	No	Yes	Yes
Recommended RAM	No	No	Yes (0.238, *0.060)****
Observations	113	113	106

*SE in parentheses. Significance levels: ***p<0.001, **p<0.01, *p<0.05*

Table 4 - ITS's coefficient estimates and SE across three models

The baseline trajectory of hardware demands before the supply shock is captured by the pre-shock slope coefficient (β_1). Across the specifications, the pre-shock slope is consistently positive, though its statistical significance varies depending on the controls applied. In Model 1, β_1 is estimated at 0.235 GB per year ($p = 0.017$). In the preferred Model 2, the coefficient increases to 0.268 GB per year and achieves statistical significance ($p = 0.004$). In the restricted Model 3, β_1 is 0.084 ($p = 0.434$). The statistically significant pre-shock slope in the preferred specification confirms that PC game minimum RAM requirements grew at a measurable, steady rate during the baseline period. This finding is entirely consistent with the Wirth's Law trajectory of continuous software bloat established in the visual analysis in Section 4.2.

The level change coefficient (β_2) measures whether there was an immediate, instantaneous shift in RAM requirements exactly at the Q3 2022 structural break point. Across all three models, the coefficients indicate an absolute jump in requirements of approximately 1.6 to 2.0 GB. Model 1 reports $\beta_2 = 2.007$ ($p = 0.218$), Model 2 reports $\beta_2 = 1.596$ ($p = 0.269$), and Model 3 reports $\beta_2 = 1.766$ ($p = 0.189$). While these represent substantive absolute increases in memory demands, none of the coefficients reach statistical significance. This indicates that while there was noticeable upward pressure on requirements exactly at the break point, the variance within the sample is too high to statistically confirm a discrete, instantaneous level jump. Therefore, rather than a sudden spike, the data suggests that the organizational response to the macroeconomic constraint manifested primarily as a trajectory shift across the years following the break. The non-significance of the individual level- and slope-change coefficients does not contradict the Chow test result. The Chow test rejects overall parameter stability across the break, whereas the segmented coefficients test for a discrete level shift and a distinct post-shock slope specifically; with only three post-shock annual observations, the analysis has limited power to isolate individual parameter, even where a joint structural difference is present.

The primary coefficient of interest for evaluating the first hypothesis is the post-shock slope change (β_3), which captures the alteration in the trajectory of minimum RAM requirements following the identified structural break. Across all estimated models, the coefficient for β_3 is consistently positive. In the basic specification of Model 1, the post-shock slope change is estimated at +0.755 ($p = 0.355$). In the preferred specification of Model 2, which incorporates genre fixed effects, the coefficient is +0.637 ($p = 0.374$). Model 3, which controls for the recommended RAM specifications, yields a coefficient of +0.357 ($p = 0.598$). Furthermore, the sensitivity check incorporating the transition year of 2022 reports a coefficient of +0.646 ($p = 0.364$). Across all specifications, β_3 is positive (the direction opposite to that predicted by H1A ($\beta_3 < 0$). Because H1 is formulated as a directional hypothesis, the observation of a positive coefficient is sufficient to reject H1A regardless of statistical significance: the data move away from the predicted region of the parameter space rather than toward it. H1 is therefore not supported and, although none of these coefficients reaches conventional thresholds for significance against a two-sided null, the directional consistency across four specifications is itself a substantive finding: minimum RAM requirements accelerated after the macroeconomic supply shock rather than slowing.

This pattern contradicts the prediction derived from the conceptual model. Several explanations are theoretically admissible and are developed in Chapter 5: (i) organisational adaptation may operate on time horizons longer than the four-year post-shock window observable here, consistent with the time-compression diseconomies and path dependencies emphasised in Chapter 2; (ii) the substitution mechanism predicted by RDT may operate through AI upscaling adoption (H2) rather than through suppression of minimum RAM, rendering H1 the wrong empirical operationalisation of the underlying theoretical claim; (iii) the supply shock may have been absorbed by manufacturers and consumers (e.g., through deferred upgrades and price-tolerated configurations) rather than transmitted to developer requirement-setting decisions during the observation window.

Evaluating the overall explanatory power of the empirical estimations, the adjusted R^2values indicate that the models successfully capture a substantial portion of the variance in minimum RAM requirements over the observation window. Model 1, which employs the basic Interrupted Time Series specification without additional controls, reports an adjusted R^2 of 0.369. The introduction of genre fixed effects in Model 2 increases the adjusted R^2 meaningfully to 0.457. Model 3 further increases the adjusted R^2 marginally to 0.533. The significant improvement in overall model fit from Model 1 to Model 2 confirms that game genre serves as a highly meaningful and necessary control variable. It demonstrates that accounting for the systematic baseline differences in computational and graphical intensity across distinct software categories is essential for accurately modeling hardware requirement trajectories.

To further explore potential heterogeneity in the organizational response across different segments of the software market, a subsample analysis was conducted. This analysis separated computationally lighter sports titles from graphically intensive titles, such as action-RPGs, to test whether adaptation strategies varied by genre. The results for the sports subsample ($N = 32$) reveal a positive post-shock slope change of $\beta_3 = +0.893$ ($p = 0.292$). The sports subsample also exhibits a minor negative level change exactly at the structural break, estimated at $\beta_2 = -0.383$ ($p = 0.816$). This suggests a slight downward shift in the baseline hardware demands for sports titles immediately at the break point, although this coefficient does not reach statistical significance. Conversely, the subsample of graphically intensive titles ($N = 61$) yields a post-shock slope change of $\beta_3 = -0.012$ ($p = 0.990$), alongside a substantial level change of $\beta_2 = +3.661$ ($p = 0.063$). Neither subsample produces statistically significant results for the primary coefficient of interest β_3 at the standard $p < 0.05$. While the expanded dataset improves overall statistical power, splitting the sample into these distinct categories (comprising 32 sports titles and 61 graphically intensive titles), the model's capacity to perfectly isolate individual parameters is still restricted. These specific subsample results, that can be found in Appendix A, should be therefore interpreted cautiously as descriptive trends.

Finally, a sensitivity check was performed to systematically verify the robustness of the primary findings against the methodological handling of the structural break boundary. As noted in the data preparation, the year 2022 was initially excluded to prevent transition-year contamination. When re-estimating the preferred specification by including 2022 as an explicit pre-shock year (expanding the sample to 125 observations), the resulting post-shock slope change is $\beta_3 = +0.646$ ($p = 0.364$). This result is nearly identical to the primary specification of Model 2 ($\beta_3 = +0.637$) in both its positive direction and its lack of statistical significance. This alignment confirms that the main findings regarding the continued acceleration of software bloat are robust to the specific 2022 boundary decision and are not artificial byproducts of the transition-year data exclusion strategy. The full results of the analysis can be found in Appendix A.

4.4 Time-Series Analysis: AI Infrastructure Demand and DRAM Price Proxy (H3)

To evaluate the macroeconomic supply shock proposed by the third hypothesis (H3), the analysis relies on a dataset of eleven annual observations spanning from 2015 to 2025. Over this observation window, the AI infrastructure demand proxy, NVIDIA Data Center revenue, experienced exponential growth, surging from $0.34 billion in 2015 to $131.42 billion in 2025, representing a 385-fold increase. In contrast, the hardware cost proxy, the semiconductor Producer Price Index (PPI), moved from 37.7 in 2015 down to 29.9 in 2025. Table 4 presents the complete annual dataset used for this estimation. As observed in the data, the PPI exhibited a long-run declining trend across most of the observation window, a trajectory consistent with historical Moore's Law cost reductions dominating the sector-wide average.

Year	***NVIDIA Data Center Revenue ($B)***	***Semiconductor PPI***
2015	0.34	37.66
2016	0.83	36.27
2017	1.93	35.74
2018	2.93	34.17
2019	2.98	32.54
2020	6.70	31.33
2021	10.61	30.01
2022	15.00	31.02
2023	47.52	31.69
2024	115.19	31.14
2025	131.42	29.86

Table 5 - NVIDIA Data Center Revenues and Semiconductor PPI (2015 - 2025)

The formal correlation analysis, that can be found in Appendix A, yields divergent results depending on the specific test applied. The Pearson correlation coefficient indicates a negative association ($r = -0.547$) that does not reach statistical significance at the $p < 0.05$ threshold ($p = 0.082$). Conversely, the Spearman rank-order correlation is both negative and highly significant ($rho = -0.891$, $p = 0.0004$). The divergence between the Pearson and Spearman results is itself informative. The significant Spearman result reflects a strong monotonic relationship: as AI infrastructure demand grew over the decade, the PPI consistently declined or stayed flat. The Pearson result is comparatively weaker because this underlying relationship is not strictly linear. It should also be noted that with only 11 annual data points, the statistical power is inherently limited for the Pearson test. H3 predicted a positive association between AI infrastructure demand and consumer DRAM prices. Because the observed correlation is negative, the null hypothesis cannot be rejected in the predicted direction.

To estimate the magnitude of this association, a simple Ordinary Least Squares (OLS) regression was conducted with the annual PPI as the dependent variable and NVIDIA Data Center revenue as the independent variable. The estimation yields an intercept of 33.79 ($p < 0.001$) and a coefficient on NVIDIA revenue of -0.031 ($p = 0.082$), with an adjusted R^2 of 0.221. The negative coefficient confirms the negative directional relationship observed in the correlation tests, though it remains statistically insignificant at the $p < 0.05$ threshold.

The failure to support H3 across the full observation window provides important empirical insight into proxy behaviour under continuous technical change. The negative overall correlation demonstrates that long-run Moore's Law cost reductions dominate the aggregated index, effectively masking the localized supply pressures. However, the measurable reversal in 2022 and 2023 provides an important structural signal. The index expansion during those specific years is temporally synchronized with the Q3 2022 structural break. This localized anomaly indicates that while the aggregate proxy is structurally dampened by sector-wide price declines, the intensity of the AI infrastructure demand shock was severe enough to briefly disrupt a decade-long macroeconomic trend.

This discrepancy highlights a fundamental measurement limitation inherent in the dataset. The FRED semiconductor PPI is a sector-wide index that cannot precisely isolate the HBM-to-consumer-DRAM crowding out effect. The negative overall correlation found in the analysis likely reflects the dominance of long-run Moore's Law cost reductions across the full 11-year observation window, which effectively masked any localized consumer DRAM price pressure during the AI-intensive period. To assess the robustness of the H3 findings against potential proxy bias, a search was conducted for alternative consumer DRAM price series, specifically DDR4 and DDR5 spot price indices, that could serve as a more granular measure of consumer memory cost volatility. However, granular consumer DRAM spot price data with sufficient historical coverage is either proprietary, as with TrendForce and DRAMeXchange subscription services, or not available in a format suitable for academic replication. Furthermore, retail consumer prices for memory modules, such as those tracked by e-commerce platforms, are subject to product generation cycles and early adoption premiums that are impertinent to the semiconductor manufacturing cost pressures this study seeks to measure, making them unsuitable as alternative proxies. The FRED semiconductor PPI therefore remains the primary proxy for hardware cost pressure in this study. This constraint is acknowledged as a data limitation and represents a clear avenue for future research when proprietary DRAM price series become accessible.

Figure 5 visually depicts the trajectories, illustrating both the historical timeline and the temporal anomaly observed during the 2022-2023 supply shock period, juxtaposed against the concurrent exponential growth in NVIDIA Data Center revenue.

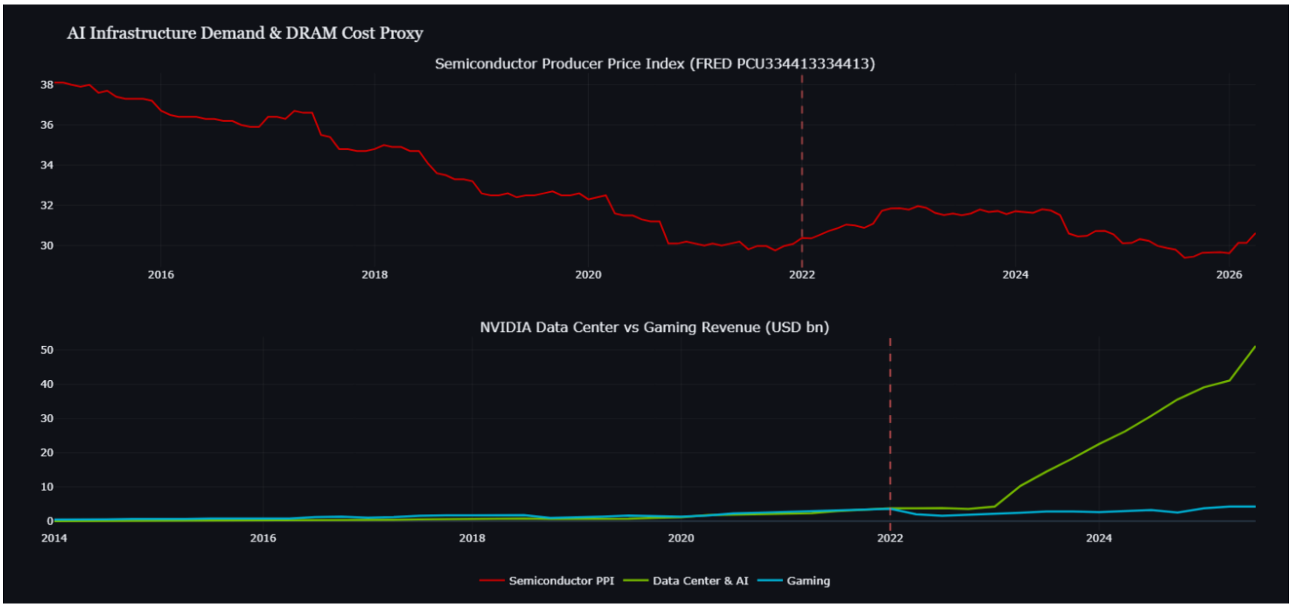

Figure 5 - Visualization NVIDIA Data Center Revenue & PPI Semiconductor

4.5 AI Upscaling Adoption Analysis (H2 and H4)

The empirical analysis of Hypothesis 2 examines whether AI upscaling adoption increased significantly during the AI-intensive period. The results provide strong support for this hypothesis.

Before introducing the formal regression models, the descriptive statistics vividly illustrate this shift. As illustrated in Figure 6, the adoption of algorithmic efficiency measures underwent a dramatic shift following the structural break. The proportion of AAA titles utilizing upscaling technologies nearly tripled, accelerating from a baseline of 22.9% in the pre-shock period to 85.4% during the AI-intensive period.

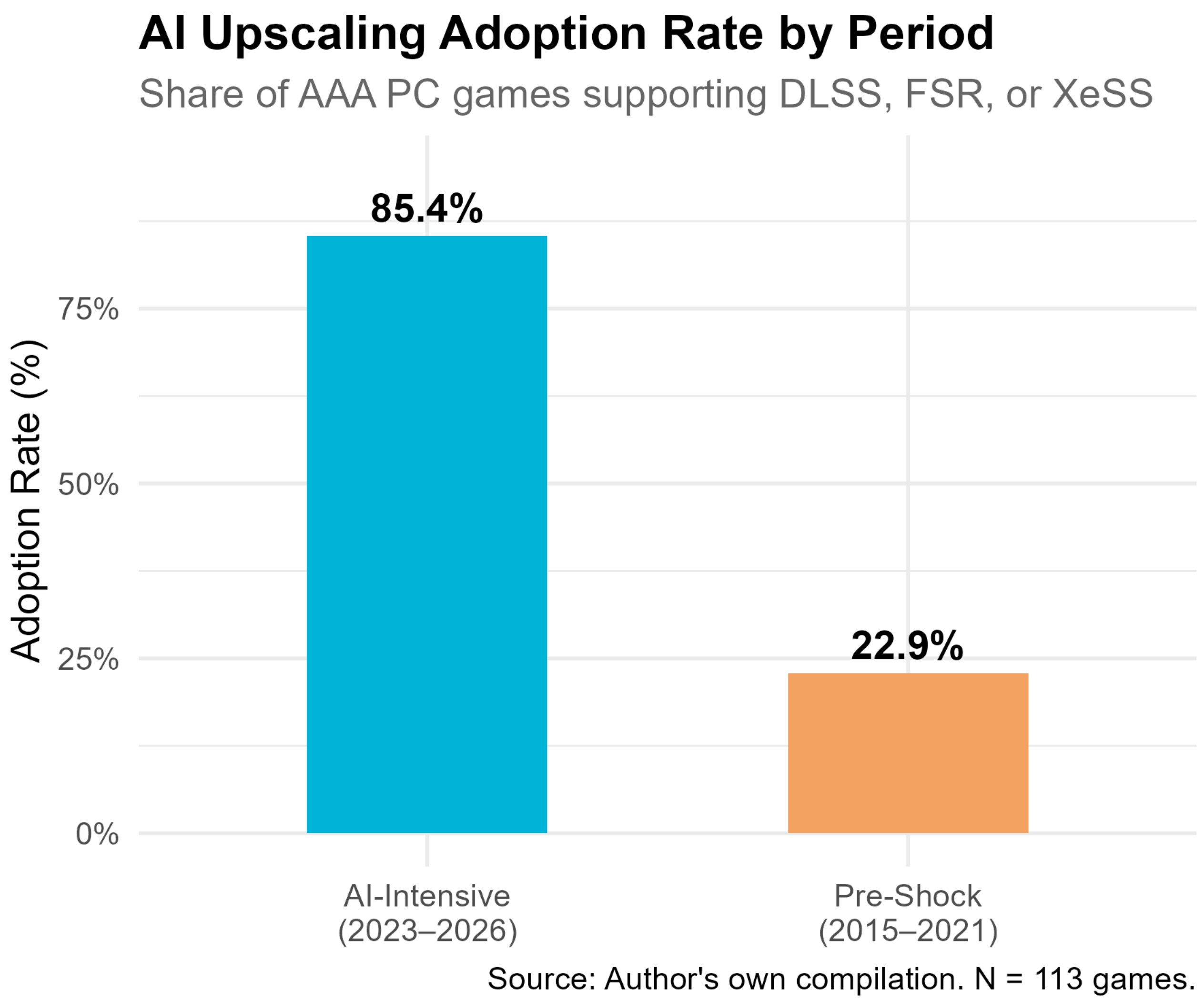

Figure 6 - AI Upscaling Adoption Rate by Period

To formally evaluate the probability of AI upscaling adoption, three distinct binomial logistic regression models were estimated. The dependent variable across all specifications is the binary integration of algorithmic upscaling technologies (DLSS/FSR). Model 1 serves as the baseline, isolating the primary effect of the post-shock period dummy. Model 2 introduces genre fixed effects to control for baseline differences in computational intensity across the software market. Finally, Model 3 incorporates a continuous time trend to rigorously isolate the discrete impact of the Q3 2022 structural break from the organic, historical diffusion of these technologies.

The primary logistic regression (Model 1) confirms this descriptive surge with high statistical significance. Evaluating AI upscaling adoption against the structural break period dummy yields a log-odds of 2.980 and an odds ratio of 19.7 ($p < 0.001$). Simply put, games released during the AI-intensive period are twenty times more likely to integrate DLSS or FSR than their pre-shock counterparts.

This core effect holds robustly in Model 2, which introduces genre fixed effects. Even when accounting for baseline differences in graphical intensity, the period dummy remains highly significant ($p < 0.001$). The genre controls reveal that computationally lighter sports titles are significantly less likely to adopt upscaling ($\beta = -3.017, p = 0.002$), while Action-RPG and FPS genres show no significant deviation from the baseline. This moderation aligns perfectly with the theoretical expectation that less demanding software faces less pressure to implement advanced optimization.

However, Model 3 introduces a necessary stipulation to the causal interpretation of this timeline. When a continuous time index is added to the specification, the specific post-shock dummy loses its statistical significance ($p = 0.159$), while the continuous time trend itself becomes significant ($\beta = 0.375, p = 0.019$). This indicates that the adoption of AI upscaling grew continuously over the observation window rather than experiencing a sudden, discrete jump exactly at the Q3 2022 structural break. This continuous trajectory reflects the historical reality of the technology's rollout: NVIDIA introduced DLSS in 2018 and AMD followed with FSR in 2021, indicating the baseline adoption curve was already accelerating prior to the specific hardware cost shock.

While Model 3 confirms that upscaling technologies were already diffusing organically through the market, this continuous trend does not invalidate the theoretical substitution mechanism. Instead, it suggests the macroeconomic supply shock acted as a critical accelerator. It transformed upscaling from a gradual technological evolution into an urgent, near-universal requirement for survival during the AI-intensive period. The evidence overwhelmingly demonstrates that the AI-intensive period is characterized by near-universal reliance on upscaling technologies, confirming the organizational substitution response predicted by Resource Dependence Theory.

	Model 1 – Basic	***Model 2 – Genre FE***	***Model 3 – Time Trend***
(Intercept)	-1.216*** (*0.285*)	-0.941* (*0.436*)	-2.675*** (*0.749*)
Post-Shock Dummy	2.980*** (*0.526*)	3.751*** (*0.798*)	1.210 (*0.859*)
Time Index	-	-	0.375* (*0.159*)
Genre: Sports	-	-3.017** (*0.957*)	-
Observation	111	111	111
Pseudo[31] R^2	0.286	0.426	0.330

**Note: Estimates represent long odds. SE in parentheses. Significance levels: ***p<0.001, **p<0.01, *p<0.05*

Table 6 - Logistic Regression Analysis of AI Upscaling Adoption

Finally, to empirically test the direct substitution mechanism proposed in Hypothesis 4, the analysis shifts from the structural break dummy to continuous macroeconomic indicators. Because granular consumer DRAM spot price data (DDR4/DDR5) is proprietary and unavailable within the study's constraints, two alternative proxies were employed:

- Specification 1 uses the FRED semiconductor PPI.
- Specification 2 uses NVIDIA Data Center revenue as a reduced-form upstream demand proxy, following the logic that AI infrastructure demand is the upstream cause of consumer DRAM supply constraints.

The logistic regression for Specification 1 yields a non-significant result for the PPI proxy ($\beta = -0.029, SE = 0.301, z = -0.095, p = 0.924,$ odds ratio = 0.97). This lack of significance is explained by the data itself. The PPI ranged only 2.6 index points across the observation window, providing insufficient statistical leverage to detect an association even if one exists. Adding genre controls and a time trend to the model did not improve the overall fit or produce significant results. Consequently, H4 is inconclusive under this specification: the proxy inadequacy prevents a valid test of the price mechanism, so the hypothesis is neither supported nor rejected.

The logistic regression for Specification 2 evaluates the NVIDIA revenue proxy. The initial model estimates a positive and highly significant relationship ($\beta = 0.030, SE = 0.009, z = 3.525, p < 0.001,$ odds ratio = 1.03 per additional billion dollars). Crucially, when a continuous time trend is introduced in Model 3, the NVIDIA revenue coefficient collapses to near zero ($\beta = 0.010, p = 0.582$). This indicates that the marginal association observed in the first model is entirely attributable to the shared time trend between AI demand growth and upscaling adoption, rather than a direct substitution relationship. Notably, the Sports genre dummy remains consistently negative and

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[31] Model fit is evaluated via McFadden's Pseudo R^2($R^2_{McFadden} = 1 - \frac{Residual\ Deviance}{Null\ Deviance}$) as standard R^2 metrics cannot be applied to logistic regression. Values between 0.2 and 0.4 denote an excellent fit.

significant across both specifications ($p \approx 0.002$ to 0.004), confirming that genre moderates technology adoption regardless of the demand proxy used. The full results of the analysis on both specifications can be found in Appendix A.

Based on these findings, H4 cannot be confirmed or rejected with the available data. Neither proxy produces a statistically significant result. The PPI specification fails due to insufficient variation in the index. The NVIDIA revenue specification confirms parallel growth trends but cannot establish a direct substitution mechanism once the underlying time trend is controlled. Testing H4 rigorously requires consumer-level DDR4 and DDR5 spot price data at a quarterly frequency, such as the commercial data available from TrendForce and DRAMeXchange, which fell outside the data collection scope of this study and represents a clear priority for future research and a limitation acknowledged in the study design.

4.6 Structural Comparison: Difference-in-Differences Specification

Before presenting the analysis, an important caveat is necessary. Because console hardware specifications are fixed at 16 GB throughout the observation window, the parallel trends assumption required for a causal difference-in-differences design is violated by construction. The following specification is therefore descriptive only: it quantifies the post-shock divergence in means between PC and console requirements, adjusted for genre. No causal interpretation is attached to the coefficient, and this analysis should not be read as a robustness check for the primary ITS results.

To illustrate the role of platform governance in transmitting the supply shock, a Difference-in-Differences specification was estimated as a structural comparison. By comparing PC games against their console counterparts, this specification measures the divergence between a hardware-fluid ecosystem and a hardware-static one. As established in the methodology, the fixed-capacity coding of console observations violates the parallel trends assumption required for strict causal inference. Consequently, the results are interpreted purely as a descriptive measure of how severely the open PC market deviated from the insulated console baseline during the AI-intensive period.

The DiD estimation, conducted on the original 2019–2026 dataset comprising 72 observations (prior to the historical extension back to 2015), yields an adjusted R^2 of 0.758. The primary DiD coefficient of interest (β_3), which captures the differential change in requirements, is estimated at +4.06 GB ($p = 0.0003$). This positive and highly statistically significant coefficient indicates that PC RAM requirements grew significantly more than console requirements during the AI-intensive period.

Crucially, both the ITS and DiD specifications tell the same directional story: RAM requirements grew after the supply shock rather than declining. While the primary ITS analysis finds a positive but non-significant post-shock slope change ($\beta_3 = 0.755, p = 0.355$), the supplementary DiD specification finds a positive and highly significant coefficient ($\beta_3 = 4.52$ GB, $p < 0.001$). Despite the methodological differences between the models and the parallel trends limitation inherent in the DiD, the consistent positive direction across both specifications suggests the finding is not an artifact of either design choice. This descriptive divergence visually aligns with the ITS findings but is not presented as causal evidence.

5. Summary, Conclusion, and Recommendations

5.1 Discussion of Findings

The empirical analysis reveals a complex and nuanced organizational response to the AI-driven supply shock. This response diverges from the predicted suppression of hardware demands but confirms a dramatic acceleration in algorithmic efficiency adoption. This overarching finding demonstrates that while the macroeconomic constraint did not halt the historical trajectory of software bloat, it fundamentally altered how development organizations manage their hardware dependencies and technical debt in response to external scarcity.

To answer the first research question (RQ1) - to what extent has AI infrastructure demand been associated with price anomalies in the consumer RAM market - the empirical evidence is suggestive but constrained by data availability. Between 2015 and 2025, NVIDIA Data Center revenue grew 385-fold, representing a scale of expansion that is unprecedented within the semiconductor industry. Concurrently, the FRED semiconductor Producer Price Index (PPI) exhibited a measurable interruption of its long-run declining trend in 2022 and 2023, which is temporally consistent with the theoretical shock period. However, the broad, sector-wide nature of the PPI prevents the precise isolation of the High-Bandwidth Memory (HBM) to consumer DRAM crowding-out effect. Consequently, Hypothesis 3 (H3) cannot be confirmed in the predicted positive direction. The Spearman correlation reveals a strong negative monotonic relationship: as AI demand grew, the sector-wide cost index declined. This reflects the dominance of historical Moore's Law cost reductions over the full observation window, which effectively masked the localized consumer DRAM price pressure documented by industry sources (TrendForce Research Team, 2026). This finding aligns with the theoretical framework established in Section 2.2 and is corroborated by recent studies documenting this exact supply reallocation and consumer market squeeze ((H. Kim, 2025a)). However, while those works rely on highly granular, proprietary spot-price data, this study demonstrates the challenge of observing these same localized shocks through public macroeconomic proxies.

To answer the second research question (RQ2) regarding how minimum RAM requirements evolved and whether the trajectory changed significantly, the analysis provides a two-part conclusion. First, the trajectory did change significantly. The Chow test confirms a structural break at Q3 2022 at $p = 0.007$ resulting in a robust finding. Second, the direction of the change was the exact opposite of the prediction formulated in H1 because RAM requirements accelerated after the shock rather than flattening. The mean minimum RAM jumped from 7.06 GB pre-shock to 11.60 GB post-shock, representing a 64.3% increase meaning that H1 is not supported. This result is theoretically interesting for two primary reasons:

1. The positive post-shock slope change coefficient for minimum RAM requirements highlights a critical reality of software development: the presence of severe time compression diseconomies and path dependency. The continued acceleration of hardware demands is not a statistical failure of the conceptual model, but rather a clear empirical demonstration of sticky asset stocks. Games released between 2023 and 2026 were already deep into their two to four year production cycles when the Q3 2022 structural break occurred. Development teams were locked into memory budgets and engine architectures established years prior. They simply could not dismantle and reconfigure these foundational assets mid-production without incurring catastrophic delays, forcing them to maintain their trajectory of software bloat despite the changing macroeconomic environment.
2. The second explanation is a market segmentation response. Rather than suppressing requirements, developers may be targeting premium hardware segments while simultaneously deploying upscaling as an accessibility tool for lower-end users. The two strategies of high RAM requirements plus upscaling support are not contradictory but complementary.

To answer the third research question (RQ3) regarding the extent to which developers substituted raw memory demands with AI optimization strategies, the data shows a strongly affirmative empirical result for the adoption behaviour but does not support the substitution mechanism as originally theorized. H2 yields the most statistically robust result in the study. Games in the AI-intensive period are twenty times more likely to support DLSS or FSR than pre-shock games. The adoption rate nearly quadrupled, accelerating from 22.9% to 85.4%. This creates a paradoxical observation: developers simultaneously increased raw memory requirements while accelerating the adoption of AI-based optimization.

This overwhelming shift toward algorithmic substitution must take in consideration the direct context of the development cycle lag discussed previously. Because organizations were caught mid-cycle with fixed and bloated RAM targets they could not easily alter, they were forced to deploy modular, low-friction optimization workarounds to survive the sudden consumer hardware price squeeze. Integrating upscaling measures provided a highly effective mechanism to service accumulated technical debt on the fly. It allowed developers to artificially lower the barrier to entry for consumers experiencing hardware constraints, all without requiring a fundamental rewrite of the core game engine. In this environment, AI upscaling functioned as a strategic adaptation deployed to compensate for the rigid architectural inertia that prevented actual RAM requirement reductions.

H4 remains inconclusive: because the available price proxy inadequately captured consumer DRAM price movements, the analysis could neither support nor reject the direct price mechanism. This reflects a data limitation rather than a theoretical refutation, and the proposed mechanism remains open for testing with a more direct price measure. The substitution behaviour nevertheless exists, as corroborated by H2. While the continuous time trend in Model 3 indicates that broader technological momentum was already underway, the data suggests the supply shock served as a critical catalyst. It forced developers to fully exploit these existing tools, accelerating their adoption from an optional performance enhancement to a mandatory organizational buffer against hardware scarcity. Within the theoretical framework established in section 2.5, this finding partially refines Resource Dependence Theory: developers responded to the external resource constraint by adopting algorithmic efficiency measures but did so alongside continuing increases in raw memory requirements rather than as a substitute for them. It is also consistent with the technical debt argument since upscaling adoption represents a form of debt paydown, substituting optimization effort for hardware requirements (Kruchten et al., 2012). Furthermore, the heterogeneous adoption rates across genres (with sports titles significantly less likely to adopt upscaling) are consistent with the dynamic capabilities framework, confirming that organizational path dependencies and asset positions constrain the speed of strategic adaptation (Eisenhardt & Martin, 2000; Teece et al., 1997).

Taken together, the empirical findings reveal that the AI-driven supply shock produced a confirmed structural break in PC game development practices. Developers did not respond by suppressing hardware requirements, instead, they responded by significantly accelerating the adoption of algorithmic efficiency tools while continuing to push hardware ambitions upward. This dual response suggests a market segmentation strategy rather than broad-based adaptation: maintaining high graphical fidelity for premium hardware users while deploying upscaling to preserve accessibility for the broader consumer base in order to avoid the shrinking of the TAM. The shock interrupted the historical Moore's Law equilibrium between hardware improvement and software bloat. The organizational response was more sophisticated and heterogeneous than a simple suppression of requirements like initially predicted.

5.2 Theoretical Implications

The theoretical framework of this study posited a fundamental dependency between software bloat (Wirth, 1995) and the hardware improvements traditionally guaranteed by Moore's Law. The primary finding regarding Hypothesis 1 offers a significant refinement to this dependency because it was initially predicted that the interruption of Moore's Law would force a suspension of Wirth's Law. However, the empirical findings suggest a more subtle reality: Wirth's Law did not suspend, as RAM requirements continued to rise. Instead, the organizational response to the supply shock, specifically the accelerated adoption of upscaling technologies, suggests that developers found an alternative mechanism to maintain the Wirth's Law trajectory while partially decoupling from raw hardware dependency. The theoretical contribution here is a critical refinement of the Moore's Law dependency argument (Leiserson et al., 2020; Xu et al., 2010). When Moore's Law fails at the hardware level, developers adopt algorithmic efficiency at the software level as a complement to, rather than a substitute for, continued growth in raw hardware requirements. Wirth's Law persisted (RAM requirements kept rising), while upscaling was layered on top as an additional coping mechanism, partially decoupling perceived performance from raw memory dependency without reducing that dependency outright. This finding suggests that future studies examining software bloat should account for the possibility of algorithmic substitution as a coping mechanism when hardware improvement cycles are disrupted.

Resource Dependence Theory (RDT) provided the theoretical framework for understanding how organizations navigate external constraints and secure necessary inputs (Pfeffer & Salancik, 2015). The robust finding for Hypothesis 2 strongly supports RDT's core prediction that organizations adapt strategically when external resources become scarce since developers did not passively accept the hardware constraints imposed by rising DRAM costs; instead, they actively reduced their structural dependency on raw consumer hardware through the aggressive adoption of upscaling technologies. This extends RDT into a novel empirical context, specifically digital entertainment software development. Historically, RDT has been used to analyze traditional supply chain relationships, but this study demonstrates how resource dependencies in digital ecosystems are uniquely mediated by platform governance structures rather than standard supplier-buyer dynamics.

The acceleration of RAM requirements combined with the explosion of upscaling adoption suggests a profound paradox: developers continued accumulating technical debt by pushing RAM requirements higher while simultaneously deploying tools to service that debt through algorithmic efficiency. This pattern is consistent with the theoretical framework where technical debt accumulation and debt paydown can occur simultaneously under different organizational pressures (Kruchten et al., 2012; Ramasubbu & Kemerer, 2016). The hardware scarcity created intense pressure to pay down debt through upscaling, while the simultaneous competitive pressure to maintain high graphical fidelity continued driving the accumulation of new debt.

The genre heterogeneity observed in the second hypothesis, where sports titles were significantly less likely to adopt upscaling, provides empirical support for the dynamic capabilities framework. Not all developers responded equally to the same external pressure. Path dependency and existing asset positions fundamentally constrain strategic pivots (Eisenhardt & Martin, 2000; Teece et al., 1997). Sports game developers, whose titles inherently possess lower graphical intensity, faced less pressure to invest in upscaling infrastructure and were consequently slower to adopt it. This heterogeneity also confirms the platform governance argument, demonstrating that the open PC ecosystem transmitted the supply shock differently across various developer segments depending heavily on their existing technical capabilities (Gawer, 2014; Tiwana et al., 2010).

5.3 Managerial Implications and Recommendations

The empirical findings of this study offer direct implications for game developers and publishers navigating an environment of hardware scarcity. The strongest finding from Hypothesis 2 reveals a 19.7 odds ratio for the adoption of AI upscaling technologies during the AI-intensive period, demonstrating near-universal integration among recent AAA titles. Developers who have not yet integrated technologies like DLSS or FSR are at a competitive disadvantage so much so tha development studios should consider treating algorithmic upscaling as a standard pipeline requirement rather than an optional feature. This is particularly critical given that the genre analysis confirms even computationally lighter titles face adoption pressure, suggesting upscaling integration is becoming an industry-wide baseline expectation rather than a premium differentiator. Furthermore, the results from Hypothesis 1 indicate that minimum RAM requirements are accelerating rather than slowing down and, with rising hardware costs threatening to price out consumers, developers must balance their hardware ambitions with accessibility tools. Releasing titles with high RAM requirements but without upscaling support increasingly shrinks the TAM, which can undermine the business value generated by the software (Brynjolfsson & Hitt, 1996; Melville et al., 2004).

For digital platform operators, such as Steam or the Epic Games Store, the macroeconomic supply shock presents a structural risk to ecosystem growth. These platforms rely on two-sided network effects, which face declining momentum if rising hardware costs cause the active consumer base to contract (Parker & Van Alstyne, 2005). To protect the health of the ecosystem, platform operators may benefit from incentivizing the adoption of AI upscaling among their third-party developers. Operators should consider implementing formal certification programs for optimized titles, offering featured storefront placement for games that meet strict accessibility criteria, or providing direct technical support and subsidized middleware for upscaling integration. By managing the technical standards of the platform, operators can help preserve the accessibility of their software library even as the underlying hardware becomes more expensive. This is particularly urgent for open PC platform operators, who unlike console manufacturers cannot enforce hardware standardization and must therefore rely on governance incentives rather than hardware mandates to maintain ecosystem accessibility.

Finally, the structural shift in semiconductor manufacturing holds broader implications for policymakers and regulatory bodies. The measurable uptick in the semiconductor Producer Price Index during 2022 and 2023, combined with the evidence of production reallocation from consumer memory to High-Bandwidth Memory (TrendForce Research Team, 2026), suggests strong downstream consumer welfare implications if AI infrastructure demand continues to crowd out the supply of consumer DRAM, the entire consumer electronics market faces sustained cost pressures. Policymakers should consider these downstream consumer welfare effects when evaluating national semiconductor manufacturing strategies and subsidy allocations by encouraging diversified fabrication capacity investments, rather than solely subsidizing cutting-edge AI accelerator facilities, they could mitigate these negative externalities and protect the affordability of general-purpose computing components for the broader public. This recommendation aligns with the broader argument that physical infrastructure constraints in AI-driven supply chains have material consequences for downstream digital ecosystems (H. Kim, 2025).

Having established that development organizations cannot simply suppress hardware requirements mid-cycle, and based on the empirical findings, this study proposes a three-pillar framework for resilient software product strategy during periods of hardware resource scarcity:

1. **Acknowledge Path Dependency:** Organizations must recognize that hardware targeting is a sticky asset because attempting to suppress memory requirements mid-production may cause catastrophic delays. Production cycles should lock in baseline targets early and avoid reactionary engine reconfigurations.
2. **Deploy Algorithmic Substitution for Technical Debt:** Instead of stripping features to meet hardware constraints, development teams should institutionalize the use of modular AI optimization tools (like DLSS/FSR) as a low-friction mechanism to pay down technical debt on the fly.
3. **Execute Hardware Market Segmentation:** Organizations should adopt a dual-targeting strategy: maintaining high-fidelity asset targets for premium hardware users, while relying heavily on upscaling APIs to artificially lower the barrier to entry for the broader consumer market facing price squeezes.

5.4 Limitations

The sample size of 125 games collected, of which 113 were used in the primary analysis following the exclusion of 12 titles released in the transition year (2022), containing only 43 post-shock titles presents a structural limitation for the analysis. This constraint is the most likely reason the coefficient β_3 in the Interrupted Time Series analysis was positive but not statistically significant and a larger and more temporally balanced sample would provide more reliable slope estimates. The post-shock window is inherently short given the recency of the structural break between 2023 and 2026 so future research utilizing a longer observation window will be better positioned to detect long-term adaptation effects.

The reliance on the FRED semiconductor Producer Price Index constitutes a clear proxy limitation. Because it is an industry-wide aggregate, it collapses diverse manufacturing sectors into a single metric, obscuring the specific capacity shifting between HBM and consumer DRAM. This lack of granularity explains the inconclusive outcomes for H3 and H4, as the localized financial pressures faced by PC consumers are smoothed out by cost reductions in other semiconductor domains. Retaining these hypotheses despite proxy limitations was necessary to demonstrate empirically that standard macro-level information systems indicators fail to capture the physical, localized resource constraints of the generative AI boom, reinforcing the eventual need for granular, proprietary datasets from sources such as TrendForce or DRAMeXchange which fell outside the financial scope of this study.

AAA games typically require development cycles spanning two to four years and titles released between 2023 and 2025 were largely already in production before the macroeconomic shock occurred. The suppression of RAM requirements predicted by Hypothesis 1 may only become visible in titles that entered production after 2023, which would appear in market releases from 2026 onwards. This study effectively captures the immediate post-shock period but not necessarily the full adaptive response of the industry.

The operationalization of AAA status through a publisher tier system introduces potential selection bias. Different publishers possess divergent technical capabilities, financial resources, and strategic priorities, meaning the sample may over-represent specific genres or development approaches favored by industry incumbents. Beyond the internal composition of the AAA segment, future research would also benefit from expanding the market scope to incorporate independent (indie) and AA titles. Because these smaller studios operate under vastly different financial and organizational constraints, examining their response to hardware price shocks would provide a valuable counterpoint to the AAA adaptation strategies documented here.

The requirement for each title to possess a qualifying console counterpart restricts the sample to cross-platform AAA releases, potentially excluding PC-exclusive titles that may exhibit distinct hardware targeting behaviors. Games developed exclusively for PC are not subject to the architectural constraints of console development and may therefore make different RAM specification decisions. This restriction improves sample homogeneity and theoretical consistency with the platform governance argument but limits the generalizability of the findings to the broader PC gaming market beyond cross-platform development contexts.

5.5 Future Research

The inability to isolate consumer DRAM spot prices with a broad sector-wide proxy limits the empirical validation of the crowding-out effect. The most direct extension of this study would be retesting H3 and H4 with proprietary consumer DDR4 and DDR5 spot price data. A quarterly price series covering 2019 to 2026 from TrendForce or DRAMeXchange would provide sufficient variation to properly test the substitution mechanism, allowing direct empirical validation of the consumer hardware squeeze.

The development cycle lag argument suggests the full organizational response to the shock will only be visible in titles entering production after 2023. Because AAA games require two to four years of development, the true adaptive response remains obscured within the current observation window. A replication of this study in 2027 or 2028 (with a sample covering releases through 2028) would capture developers who made specification decisions knowing the post-shock hardware environment, representing the most important direct extension of the current study.

A significant methodological extension would become possible if game-specific console RAM allocation data were to become publicly available. Currently, console observations are constrained to the hardware ceiling of 16GB by design, as no equivalent to PCGamingWiki exists for documenting game-level memory usage on PlayStation or Xbox platforms broadly. If such data became accessible, the parallel trends assumption of a Difference-in-Differencess design could be properly evaluated with genuine variation on both sides of the comparison. A properly specified DiD, comparing PC game RAM requirements against console counterparts with observed rather than imputed memory allocations, would provide stronger causal identification of the supply shock effect than the ITS design employed here, and represents a natural methodological extension of this study.

The quantitative findings raise questions about the organizational decision-making process behind RAM specification choices. It is unclear whether the development cycle lag explanation or the market segmentation explanation better accounts for the H1 null result. Qualitative research, ideally through interviews with technical directors, producers, or engine architects at major studios, would illuminate these internal mechanisms. A mixed methods follow-up would significantly strengthen the overall theoretical contribution of this work.

5.6 Conclusion

This study set out to examine how the AI-driven reallocation of semiconductor capacity toward High-Bandwidth Memory affected PC game developers who had historically depended on continuously improving and cheapening consumer hardware. The empirical analysis identifies a statistically significant structural break in PC game RAM requirements at Q3 2022 and documents a dramatic acceleration in AI upscaling adoption, with post-shock titles 20 times more likely to support DLSS or FSR than their pre-shock counterparts. The positive post-shock trajectory reflects the profound structural inertia inherent in AAA game development, where path-dependent production timelines prevent rapid adjustments to baseline hardware targets. Consequently, organizational adaptation manifests not through the immediate suppression of physical requirements, but through the accelerated deployment of software-level optimization to manage technical debt on the fly. Together, these findings suggest that when Moore's Law fails at the hardware level, organizational adaptation occurs at the software level through the complementary adoption of algorithmic efficiency measures, rather than through the substitution or suppression of hardware requirements that the theory initially predicted. This nuanced response refines the theoretical relationship between Wirth's Law and Moore's Law dependency. As AI infrastructure demand continues to reshape semiconductor manufacturing priorities, the theoretical framework developed in this study may extend beyond gaming to any software-dependent industry facing hardware resource constraints in an era of accelerating AI investment. The organizational response to hardware scarcity is not passive acceptance: it is strategic adaptation through algorithmic innovation.

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Appendix

Appendix A – Supplementary Statistical Outputs

The following appendix provides the unedited statistical outputs generated via the R environment during the data analysis phase of this study. These comprehensive logs are included to ensure methodological transparency and to permit the independent verification of the coefficient estimates, standard errors, and model fit statistics reported in the main text.

Contents:

Structural Break Estimation (Chow Test) [Tests H1]: Analysis of variance (ANOVA) output confirming the statistically significant structural break in the dataset at Q3 2022. This provides the foundational validation required to test Hypothesis 1 regarding the macroeconomic supply shock.

```
Analysis of Variance Table

Model 1: min_ram_gb ~ time_index
Model 2: min_ram_gb ~ time_index * break_dummy

  Res.Df     RSS Df Sum of Sq      F   Pr(>F)
1    111 1047.64
2    109  956.29  2    91.352 5.2063 0.006927 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Interrupted Time Series Analysis [Tests H1]: Ordinary Least Squares (OLS) regression summaries (Models 1–3) evaluating the impact of the hardware supply shock on minimum RAM requirements. These models directly test the primary substitution versus complementarity effects proposed in Hypothesis 1.

```
Model 1 (Basic ITS)
OLS estimation, Dep. Var.: min_ram_gb
Observations: 113
Standard-errors: Heteroskedasticity-robust

                 Estimate Std. Error   t value  Pr(>|t|)
(Intercept)      6.252724   0.348462 17.943781 < 2.2e-16 ***
time_index       0.234622   0.097169  2.414577  0.017418 *
post             2.007230   1.620242  1.238846  0.218064
time_since_break 0.754721   0.812726  0.928629  0.355133
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.90908   Adj. R2: 0.368633
```

```
Model 2 (ITS with Genre FE)
OLS estimation, Dep. Var.: min_ram_gb
```

```
Observations: 113
Fixed-effects: genre: 4
Standard-errors: Heteroskedasticity-robust

                  Estimate Std. Error  t value Pr(>|t|)
time_index        0.268012   0.091279 2.936164 0.004076 **
post              1.596185   1.437295 1.110548 0.269276
time_since_break 0.637101   0.714133 0.892132 0.374343
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.66162     Adj. R2: 0.456518
                Within R2: 0.363045
```

```
Model 3 (ITS with Genre FE + Recommended RAM)
OLS estimation, Dep. Var.: min_ram_gb
Observations: 106
Fixed-effects: genre: 4
Standard-errors: Heteroskedasticity-robust

                  Estimate Std. Error  t value   Pr(>|t|)
time_index        0.084356   0.107420 0.785289 0.43417840
post              1.765796   1.334467 1.323223 0.18884029
time_since_break 0.356849   0.673881 0.529542 0.59762687
rec_ram_gb        0.237759   0.059601 3.989155 0.00012793 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.44061     Adj. R2: 0.532855
                Within R2: 0.464321
```

Logistic Regression Analysis [Tests H2]: Generalized Linear Model (GLM) summaries evaluating the adoption probability of AI upscaling technologies (DLSS, FSR, XeSS). This explicitly tests Hypothesis 2 regarding developer adaptation and software-level optimization, providing the basis for the McFadden's Pseudo R^2calculations.

```
Model 1 – Basic (post only):
Call: glm(formula = has_upscaling ~ post, family = binomial, data = upscaling)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.2164     0.2846  -4.273 1.92e-05 ***
post          2.9800     0.5256   5.670 1.43e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

    Null deviance: 153.15  on 110  degrees of freedom
Residual deviance: 109.39  on 109  degrees of freedom
AIC: 113.39
```

```
Model 2 - With genre controls:
Call: glm(formula = has_upscaling ~ post + genre, family = binomial,
    data = upscaling)

Coefficients:
```

```
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -0.9407     0.4364  -2.155  0.03113 *
post              3.7507     0.7983   4.698 2.63e-06 ***
genreAction-RPG   0.2685     0.7748   0.347  0.72893
genreFPS          0.4684     0.6636   0.706  0.48025
genreSports      -3.0170     0.9573  -3.152  0.00162 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

    Null deviance: 153.148  on 110  degrees of freedom
Residual deviance:  87.839  on 106  degrees of freedom
AIC: 97.839
```

```
Model 3 - With time trend:
Call: glm(formula = has_upscaling ~ time_index + post, family = binomial,
    data = upscaling)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.6745     0.7494  -3.569 0.000359 ***
time_index    0.3749     0.1592   2.355 0.018523 *
post          1.2096     0.8585   1.409 0.158866
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

    Null deviance: 153.15  on 110  degrees of freedom
Residual deviance: 102.68  on 108  degrees of freedom
AIC: 108.68
```

Sensitivity Analysis [Checks H1]: Alternative OLS specification retaining the transitional year (2022). This serves as a methodological safeguard to verify the robustness of the primary ITS coefficients and validate the core findings of Hypothesis 1.

```
OLS estimation, Dep. Var.: min_ram_gb

Observations: 125
Fixed-effects: genre: 4
Standard-errors: Heteroskedasticity-robust

                 Estimate Std. Error  t value   Pr(>|t|)
time_index       0.256747   0.069960 3.669882 0.00036568 ***
post             1.647859   1.398629 1.178196 0.24108904
time_since_break 0.645967   0.709415 0.910564 0.36438166
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.56257     Adj. R2: 0.459263
                Within R2: 0.360787
```

Correlation Analysis [Tests H3]: Pearson and Spearman correlation tests, alongside an OLS regression, evaluating the relationship between the broader semiconductor Producer Price Index (PPI) and NVIDIA Data Center revenue. This explicitly tests Hypothesis 3 regarding the divergence of AI demand from general semiconductor manufacturing costs.

```
Pearson's product-moment correlation

data:  h3_df$dc_revenue and h3_df$ppi_value
t = -1.9584, df = 9, p-value = 0.08186
sample estimates: cor = -0.5466318
```

```
Spearman's rank correlation rho

data:  h3_df$dc_revenue and h3_df$ppi_value
S = 416, p-value = 0.0004284
sample estimates: rho = -0.8909091
```

```
OLS: PPI ~ NVIDIA Data Center Revenue:

Call: lm(formula = ppi_value ~ dc_revenue, data = h3_df)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 33.78855    0.85842  39.361 2.19e-11 ***
dc_revenue  -0.03060    0.01562  -1.958   0.0819 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.368 on 9 degrees of freedom
Multiple R-squared:  0.2988,   Adjusted R-squared:  0.2209
```

Macroeconomic Demand Proxy Analysis [Tests H4]: Logistic regression models (GLM) evaluating the adoption probability of AI upscaling against the semiconductor PPI and NVIDIA Data Center revenue proxies. This tests Hypothesis 4 regarding how specific macroeconomic cost indicators drive software optimization behavior.

```
--- PART A: DRAM Price Proxy (PPI) ---
```

```
Model 1 - Basic (upscaling ~ PPI):

Call: glm(formula = has_upscaling ~ ppi_value, family = binomial, data = h4_df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.36139    9.34506   0.146    0.884
ppi_value   -0.02868    0.30058  -0.095    0.924
---
AIC: 107.93
```

```
Model 2 - With genre controls:

Call: glm(formula = has_upscaling ~ ppi_value + genre, family = binomial,
    data = h4_df)
```

```
Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      2.03173   10.85496   0.187  0.85153
ppi_value       -0.03913    0.34758  -0.113  0.91037
genreAction-RPG  1.32540    0.86049   1.540  0.12349
genreFPS        -0.23945    0.71613  -0.334  0.73810
genreSports     -2.13659    0.70624  -3.025  0.00248 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
---
AIC: 92.677
```

```
Model 3 - With time trend:

Call: glm(formula = has_upscaling ~ ppi_value + time_index, family = binomial,
    data = h4_df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -13.1384    11.0273  -1.191    0.233
ppi_value     0.2860     0.3487   0.820    0.412
time_index    0.6945     0.1636   4.246 2.18e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
---
AIC: 86.987
```

```
--- PART B: AI Demand Proxy (NVIDIA Data Center Revenue) ---
```

```
Model 1 - Basic (upscaling ~ NVIDIA DC Revenue):

Call: glm(formula = has_upscaling ~ dc_revenue, family = binomial,
    data = h4b_df)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.676540   0.357086  -1.895 0.058143 .
dc_revenue   0.030056   0.008525   3.525 0.000423 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
---
AIC: 86.963
```

```
Model 2 - With genre controls:

Call: glm(formula = has_upscaling ~ dc_revenue + genre, family = binomial,
    data = h4b_df)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)     -0.27337    0.52420  -0.521  0.60202
dc_revenue       0.03411    0.01076   3.171  0.00152 **
genreAction-RPG  0.64543    0.95266   0.678  0.49809
genreFPS         0.13946    0.76671   0.182  0.85567
genreSports     -2.64820    0.91286  -2.901  0.00372 **
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
---
AIC: 75.778
```

```
Model 3 - With time trend:

Call: glm(formula = has_upscaling ~ dc_revenue + time_index, family = binomial,
    data = h4b_df)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.196944   2.062840  -1.550    0.121
dc_revenue   0.009538   0.017341   0.550    0.582
time_index   0.484880   0.387474   1.251    0.211
---
AIC: 87.346
```

Subsample Heterogeneity Analysis [Checks H1]: Interrupted Time Series (ITS) regression models evaluating the structural break in minimum RAM requirements across distinct genre subsamples (Sports vs. Graphically Intensive). This tests whether adaptation strategies to the Q3 2022 macroeconomic shock varied significantly by genre, further contextualizing the primary findings of Hypothesis 1.

```
--- PART A: Sport Subsample (N=32) ---
```

```
Model 1 – Sports ITS:

OLS estimation, Dep. Var.: min_ram_gb
Observations: 32
Standard-errors: Heteroskedasticity-robust

                 Estimate Std. Error   t value   Pr(>|t|)
(Intercept)      5.420732   0.812510  6.671585 3.0647e-07 ***
time_index       0.189024   0.205997  0.917610 3.6666e-01
post            -0.383247   1.631319 -0.234930 8.1597e-01
time_since_break  0.892943   0.830711  1.074915 2.9159e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.21558   Adj. R2: 0.121661
```

```
--- PART B: Graphically Intensive Subsample (N=61) ---
```

```
Model 2 – Intensive ITS:

OLS estimation, Dep. Var.: min_ram_gb
Observations: 61
Standard-errors: Heteroskedasticity-robust

Estimate Std. Error   t value  Pr(>|t|)
(Intercept)      6.325938   0.389120 16.257033 < 2.2e-16 ***
time_index       0.318889   0.125712  2.536665  0.013954 *
post             3.660965   1.929673  1.897194  0.062872 .
```

```
time_since_break -0.012116   0.953016 -0.012713  0.989901
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 2.96214   Adj. R2: 0.438515
```

Appendix B – Final Sample Master Dataset

The following table presents the complete dataset of 125 AAA cross-platform PC games utilized for the Interrupted Time Series and logistic regression analyses. The data, compiled via the Steam Store API and manual verification, tracks hardware memory constraints and optimization strategies between 2015 and 2026.

Data Dictionary:

- **Title:** The official release title of the game. Note: Special characters (such as colons) have been explicitly eliminated from these strings to ensure clean data parsing and to prevent syntax errors during the R-based statistical analysis.
- **Release Date:** The exact initial launch date of the title on the PC platform, standardized to a uniform format.
- **Genre:** The primary categorical classification of the title (e.g., Action-RPG, FPS, Sports), which serves as the moderating variable in the regression models.
- **Steam App ID:** The unique numerical identifier assigned to the software on the Steam storefront. This was utilized to ensure precise data retrieval and accurate cross-referencing during the scraping process.
- **Min RAM (GB) / Rec RAM (GB):** The minimum and recommended Random Access Memory requirements at launch.
- **Upscaling (DLSS / FSR / XeSS):** Binary variables indicating official support for specific AI-driven upscaling technologies. For readability in this printed appendix, values have been transformed to "Yes" and "No". The raw dataset hosted on GitHub retains the original binary format (1 = Yes, 0 = No).

Title	*Date of release*	*Genre*	*Min RAM (GB)*	*Rec RAM(GB)*	*STEAM ID*	*Upscaling*	*DLSS*	*FSR*	*XESS*
The Witcher 3	18/05/2015	Action-RPG	6,0	8,0	292030	Yes	Yes	No	No
Batman Arkham Knight	23/06/2015	Action-Adventure	6,0	8,0	208650	No	No	No	No
F1 2015	09/07/2015	Sports	4,0	8,0	286570	No	No	No	No
Metal Gear Solid V	01/09/2015	Action-Adventure	4,0	8,0	287700	No	No	No	No
NBA 2K16	29/09/2015	Sports	4,0	0,0	329010	No	No	No	No
Call of Duty Black Ops III	05/11/2015	FPS	6,0	0,0	311210	No	No	No	No
Fallout 4	09/11/2015	Action-RPG	8,0	8,0	377160	No	No	No	No
Assassin's Creed Syndicate	18/11/2015	Action-Adventure	6,0	8,0	368500	No	No	No	No
Just Cause 3	30/11/2015	Action-Adventure	8,0	8,0	225540	No	No	No	No
Rainbow Six Siege	01/12/2015	FPS	8,0	8,0	359550	Yes	Yes	No	No
Tom Clancy's Division 2	07/03/2016	Action-Adventure	6,0	8,0	365590	No	No	No	No
Deus Ex Mankind Divided	23/08/2016	Action-RPG	8,0	16,0	337000	No	No	No	No
F1 2016	07/09/2016	Sports	8,0	8,0	391040	No	No	No	No
NBA 2K17	20/09/2016	Sports	4,0	8,0	385760	No	No	No	No
Mafia III	07/10/2016	Action-Adventure	6,0	8,0	360430	No	No	No	No
Call of Duty Infinite Warfare	03/11/2016	FPS	8,0	0,0	292730	No	No	No	No
Dishonored 2	11/11/2016	Action-Adventure	8,0	16,0	403640	No	No	No	No
Watch Dogs 2	28/11/2016	Action-Adventure	6,0	8,0	447040	No	No	No	No
NBA 2K18	02/02/2017	Sports	4,0	8,0	560820	No	No	No	No
F1 2017	24/08/2017	Sports	8,0	8,0	515220	No	No	No	No
FIFA 18	29/09/2017	Sports	8,0	16,0	1237980	No	No	No	No
Middle Earth Shadow of War	09/10/2017	Action-RPG	6,0	12,0	356190	No	No	No	No
Assassin's Creed Origins	26/10/2017	Action-Adventure	6,0	8,0	582160	No	No	No	No
Wolfenstein II	26/10/2017	FPS	8,0	16,0	612880	No	No	No	No
Call of Duty WWII	02/11/2017	FPS	8,0	0,0	476600	No	No	No	No
Need for Speed Payback	10/11/2017	Sports	6,0	8,0	621490	No	No	No	No
Far Cry 5	26/03/2018	FPS	8,0	8,0	552520	No	No	No	No
Monster Hunter World	08/08/2018	Action-RPG	8,0	8,0	582010	No	No	No	No
F1 2018	23/08/2018	Sports	8,0	8,0	737800	No	No	No	No
NBA 2K19	11/09/2018	Sports	4,0	8,0	652670	No	No	No	No
Shadow of the Tomb Raider	14/09/2018	Action-Adventure	8,0	16,0	750920	Yes	Yes	No	No
FIFA 19	28/09/2018	Sports	8,0	16,0	1238840	No	No	No	No
Assassin's Creed Odyssey	05/10/2018	Action-Adventure	8,0	8,0	812140	No	No	No	No
Devil May Cry 5	07/03/2019	Action-Adventure	8,0	8,0	601150	No	No	No	No
Sekiro Shadows Die Twice	21/03/2019	Action-Adventure	4,0	8,0	814380	No	No	No	No
Assetto Corsa Competizione	29/05/2019	Sports	4,0	16,0	805550	Yes	No	Yes	No
Battlefield V	20/11/2019	FPS	8,0	12,0	1260630	Yes	Yes	No	No
Red Dead Redemption 2	05/12/2019	Action-Adventure	8,0	12,0	1174180	No	No	No	No
Metro Exodus	14/02/2020	FPS	8,0	8,0	412020	Yes	Yes	No	No
Borderlands 3	13/03/2020	FPS	6,0	16,0	397540	No	No	No	No
Doom Eternal	19/03/2020	FPS	8,0	8,0	782330	No	No	No	No
Resident Evil 3	02/04/2020	Action-Adventure	8,0	8,0	952060	No	No	No	No
NBA 2K21	17/05/2020	Sports	4,0	8,0	1320010	No	No	No	No
Immortals Fenyx Rising	04/06/2020	Action-Adventure	8,0	16,0	1222680	No	No	No	No
FIFA 21	11/06/2020	Sports	8,0	4,0	1238820	No	No	No	No
F1 2020	10/07/2020	Sports	8,0	16,0	1080110	No	No	No	No
Death Stranding	14/07/2020	Action-Adventure	8,0	8,0	1190460	Yes	Yes	No	No
FIFA 20	07/08/2020	Sports	4,0	8,0	1255690	No	No	No	No
Control	27/08/2020	Action-Adventure	8,0	16,0	870780	Yes	Yes	No	No
Mafia Definitive Edition	24/09/2020	Action-Adventure	6,0	16,0	1030840	No	No	No	No
FIFA 17	14/10/2020	Sports	8,0	8,0	1343400	No	No	No	No

Title	*Date of release*	*Genre*	*Min RAM (GB)*	*Rec RAM(GB)*	*STEAM ID*	*Upscaling*	*DLSS*	*FSR*	*XESS*
The Outer Worlds	23/10/2020	Action-RPG	4,0	8,0	578650	No	No	No	No
Watch Dogs Legion	29/10/2020	Action-Adventure	8,0	8,0	2089820	Yes	Yes	No	No
Godfall	12/11/2020	Action-RPG	12,0	16,0	928940	No	No	No	No
Cyberpunk 2077	09/12/2020	Action-RPG	12,0	16,0	1091500	Yes	Yes	Yes	No
Outriders	01/04/2021	Action-Adventure	8,0	16,0	680420	No	No	No	No
FIFA 16	13/04/2021	Sports	4,0	8,0	1325910	No	No	No	No
Resident Evil Village	06/05/2021	Action-Adventure	8,0	16,0	1196590	Yes	Yes	No	No
Days Gone	17/05/2021	Action-Adventure	8,0	16,0	1259420	Yes	No	Yes	No
F1 2021	15/07/2021	Sports	8,0	16,0	1134570	No	No	No	No
NBA 2K22	21/07/2021	Sports	4,0	8,0	1488740	No	No	No	No
Deathloop	13/09/2021	FPS	12,0	16,0	1252330	No	No	No	No
Fifa 22	01/10/2021	Sports	8,0	0,0	1811700	No	No	No	No
Back 4 Blood	12/10/2021	FPS	8,0	12,0	924970	No	No	No	No
Far Cry 6	19/10/2021	FPS	8,0	16,0	1681320	Yes	Yes	Yes	No
Marvel's Guardians of the Galaxy	26/10/2021	Action-Adventure	8,0	16,0	1088850	No	No	No	No
Forza Horizon 5	08/11/2021	Sports	8,0	16,0	1551360	Yes	Yes	No	No
Halo Infinite	15/11/2021	FPS	8,0	16,0	1240440	No	No	No	No
Battlefield 2042	19/11/2021	FPS	8,0	16,0	1517290	Yes	Yes	No	No
Call of Duty Vanguard	21/11/2021	FPS	6,0	12,0	1085660	Yes	Yes	Yes	No
God of War	14/01/2022	Action-Adventure	8,0	8,0	1593500	Yes	Yes	No	No
Hitman 3	20/01/2022	Action-Adventure	8,0	16,0	1659040	No	No	No	No
Dying Light 2	03/02/2022	Action-Adventure	8,0	16,0	534380	Yes	Yes	Yes	No
Elden Ring	24/02/2022	Action-RPG	12,0	16,0	1245620	Yes	Yes	No	No
A Plague Tale Requiem	14/04/2022	Action-Adventure	8,0	16,0	1952490	Yes	Yes	Yes	No
FF7 Remake Intergrade	17/06/2022	Action-RPG	8,0	12,0	1462040	Yes	Yes	No	No
F1 22	01/07/2022	Sports	8,0	16,0	1692250	Yes	Yes	Yes	No
Marvel's Spider-Man Remastered	12/08/2022	Action-Adventure	8,0	16,0	1817070	Yes	Yes	No	No
NBA 2K23	08/09/2022	Sports	4,0	8,0	2273450	No	No	No	No
FIFA 23	30/09/2022	Sports	8,0	12,0	1811710	No	No	No	No
Uncharted Legacy of Thieves	19/10/2022	Action-Adventure	8,0	16,0	1659420	Yes	Yes	No	No
Assassin's Creed Valhalla	06/12/2022	Action-Adventure	8,0	8,0	2208920	Yes	No	Yes	No
Forspoken	24/01/2023	Action-RPG	16,0	24,0	1952091	Yes	Yes	Yes	No
Dead Space Remake	27/01/2023	Action-Adventure	16,0	16,0	1693980	Yes	Yes	Yes	No
Hogwarts Legacy	10/02/2023	Action-RPG	16,0	16,0	990080	Yes	Yes	Yes	No
WWE 2K24	10/02/2023	Sports	4,0	0,0	2285950	No	No	No	No
Atomic Heart	20/02/2023	FPS	8,0	16,0	668580	Yes	Yes	Yes	No
Resident Evil 4 Remake	23/03/2023	Action-Adventure	8,0	16,0	2050650	Yes	Yes	Yes	No
The Last of Us Part I	28/03/2023	Action-Adventure	16,0	16,0	1888930	Yes	Yes	Yes	No
Silent Hill 2 Remake	28/03/2023	Action-Adventure	8,0	10,0	2138710	Yes	Yes	Yes	No
EA Sports PGA Tour	07/04/2023	Sports	8,0	12,0	1796010	No	No	No	No
F1 23	15/06/2023	Sports	8,0	16,0	2108330	Yes	Yes	Yes	No
Remnant 2	25/07/2023	Action-RPG	16,0	16,0	1282100	Yes	Yes	Yes	No
Baldur's Gate 3	03/08/2023	Action-RPG	8,0	16,0	1086940	Yes	No	Yes	No
Starfield	05/09/2023	Action-RPG	16,0	16,0	1716740	Yes	Yes	Yes	No
NBA 2K24	07/09/2023	Sports	6,0	8,0	2338770	No	No	No	No
Lies of P	18/09/2023	Action-RPG	8,0	16,0	1627720	Yes	No	Yes	No
FIFA 24	28/09/2023	Sports	8,0	12,0	2195250	No	No	No	No
Assassin's Creed Mirage	05/10/2023	Action-Adventure	8,0	16,0	2620590	Yes	Yes	Yes	Yes
Lords of the Fallen	13/10/2023	Action-RPG	12,0	16,0	1501750	Yes	Yes	Yes	No
Call of Duty MW3 2023	10/11/2023	FPS	8,0	16,0	2519060	Yes	Yes	Yes	No
Avatar Frontiers of Pandora	07/12/2023	Action-Adventure	16,0	0,0	1898069	Yes	Yes	Yes	No

Title	*Date of release*	*Genre*	*Min RAM (GB)*	*Rec RAM(GB)*	*STEAM ID*	*Upscaling*	*DLSS*	*FSR*	*XESS*
Like a Dragon Infinite Wealth	25/01/2024	Action-RPG	8,0	16,0	2072450	Yes	No	Yes	No
Star Wars Outlaws	30/01/2024	Action-Adventure	16,0	6,0	2803270	Yes	Yes	Yes	No
Dragon's Dogma 2	21/03/2024	Action-RPG	16,0	16,0	2054970	Yes	Yes	Yes	No
Ghost of Tsushima	16/05/2024	Action-Adventure	8,0	16,0	2215430	Yes	Yes	Yes	No
F1 24	31/05/2024	Sports	8,0	16,0	2488060	Yes	Yes	Yes	No
Black Myth: Wukong	19/08/2024	Action-RPG	16,0	16,0	2358720	Yes	Yes	Yes	No
Warhammer 40K Space Marine 2	09/09/2024	Action-Adventure	8,0	16,0	2183900	Yes	Yes	Yes	No
EA FC 25	27/09/2024	Sports	8,0	12,0	2731540	No	No	No	No
Metaphor ReFantazio	10/10/2024	Action-RPG	6,0	8,0	2679460	Yes	No	Yes	No
Dragon Age Veilguard	31/10/2024	Action-RPG	16,0	16,0	1845910	Yes	Yes	Yes	No
Stalker 2	20/11/2024	FPS	16,0	32,0	1643320	Yes	Yes	Yes	No
NBA 2K25	05/12/2024	Sports	16,0	16,0	2767030	No	No	No	No
Indiana Jones Great Circle	08/12/2024	Action-Adventure	16,0	32,0	2677660	Yes	Yes	Yes	No
Alan Wake 2	19/12/2024	Action-Adventure	16,0	24,0	1850050	Yes	Yes	Yes	No
FF7 Rebirth	23/01/2025	Action-RPG	16,0	16,0	2909400	Yes	Yes	Yes	No
Marvel's Spider-Man 2	30/01/2025	Action-Adventure	16,0	16,0	2119490	Yes	Yes	Yes	No
Kingdom Come Deliverance 2	04/02/2025	Action-RPG	16,0	32,0	1702540	Yes	Yes	Yes	No
Monster Hunter Wilds	27/02/2025	Action-RPG	16,0	16,0	2246340	Yes	Yes	Yes	No
Assassin's Creed Shadows	20/03/2025	Action-Adventure	8,0	12,0	2933620	Yes	Yes	Yes	Yes
Clair Obscur Expedition 33	22/04/2025	Action-RPG	8,0	16,0	2486920	Yes	Yes	Yes	No
Doom The Dark Ages	15/05/2025	FPS	16,0	32,0	2413590	Yes	Yes	Yes	No
NBA 2K26	05/09/2025	Sports	8,0	16,0	3472040	No	No	No	No
EA Sports FC 26	26/09/2025	Sports	8,0	16,0	3405690	No	No	No	No

Table 7 - Final Sample Master Dataset

Appendix C – Additional dashboard

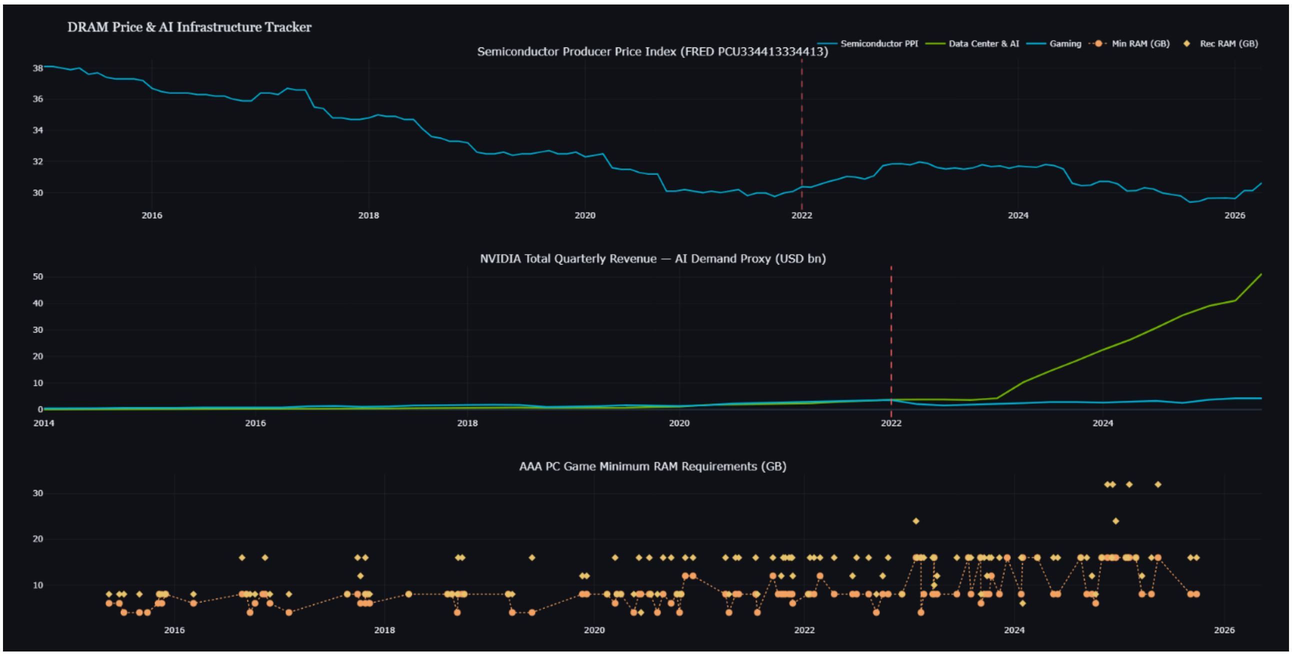

Figure 7 – Semiconductor PPI, AI Demand Proxy, and AAA PC Game Requirements (2015 - 2026)

Appendix D – Usage log for External Tools

D.1 Technology Statement

During the preparation of this work, I used ChatGPT, Perplexity and Claude to check spelling, grammar, enhance readability, brainstorm, rephrase and support in the development of my code. The following parts of the assignment were affected by AI tool usage: Literature Review, Methodology, Results and Limitations. After using this tool, Salvatore Caldara evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Salvatore Caldara takes full responsibility for the content of their work.

D.2 Logbook for the usage of AI Tools

Instance of AI Usage:

Research Stage: Writing & Editing (across the chapters)

Purpose of AI Use and Tool Used: Claude Opus 4.6 and Claude Opus 4.7 (Anthropic) were used to rephrase and refine sections of the thesis where the intended argument was clear, but the phrasing was either too informal, unclear, or inconsistent with the academic register of the document. The underlying content, ideas, and reasoning remained the author's own.

Critical Reflection and Responsible Usage: Each reformulation was evaluated against the original intent to ensure the rephrasing accurately reflected the author's argument and did not introduce new claims or alter the meaning. The prompt used was the following:

"You are an expert academic editor. Please review the following draft text for a Master's thesis in Information Management. Your task is to refine the phrasing to meet a highly formal academic register, improve clarity, and correct any grammatical or structural inconsistencies.

Strict Constraints:

- *Do not alter the underlying argument or logical flow.*
- *Do not introduce any new claims, concepts, or theoretical frameworks.*
- *Do not add any external information or expand on the text.*
- *Preserve my original meaning and intent exactly as written; only refine the vocabulary and syntax.*
- *Do not add sentences.*

Here is the text to refine: [Insert Text]"

Instance of AI Usage:

Research Stage: Literature Review

Purpose of AI Use and Tool Used: Perplexity Sonar 2 (Perplexity) was used to identify relevant academic journals and generate targeted search prompts to locate valid papers at the intersection of Information Economics and Software Engineering.

Critical Reflection and Responsible Usage: To mitigate the risk of hallucinated references, Perplexity was never used to retrieve papers directly or generate citations. All identified prompts were independently verified and specifically selected from Google Scholar after reading and understanding them in first person.

Instance of AI Usage:

Research Stage: Literature Review

Purpose of AI Use and Tool Used: Copilot (Microsoft) was used to help schematize and summarize the structure of academic papers, facilitating a more efficient review process.

Critical Reflection and Responsible Usage: Copilot's summaries were used purely as a navigational aid and were never substituted for direct engagement with the source material. All claims and citations were verified through first-hand reading of the original papers.

Instance of AI Usage:

Research Stage: Archival Data Collection

Purpose of AI Use and Tool Used: ChatGPT 4.5 (OpenAI) was used to gain a preliminary understanding of how API data retrieval works, specifically in the context of extracting data from SteamDB, FRED, and IGDB for the purposes of this study.

Critical Reflection and Responsible Usage: ChatGPT was used exclusively as a learning aid to understand the technical process, not to write or execute any code. All methodological decisions regarding data collection remained the author’s own.

Instance of AI Usage:

Research Stage: Manuscript Review and Quality Assurance

Purpose of AI Use and Tool Used: Claude Opus 4.8 (Anthropic) was provided with the drafted thesis manuscript and prompted to act as a critical academic reviewer to identify potential logical gaps, methodological inconsistencies, or unsupported claims within the research narrative.

Critical Reflection and Responsible Usage: The AI's feedback was utilized purely as a diagnostic tool to pressure-test the coherence of the arguments. All highlighted vulnerabilities were critically evaluated, and any subsequent structural refinements, theoretical clarifications, or methodological defences were independently authored and implemented to ensure the integrity of the final research.

Instance of AI Usage:

Research Stage: Coding Support

Purpose of AI Use and Tool Used: Claude Code (Anthropic) was used as an interactive programming assistant to help troubleshoot errors and optimize the Python data collection scripts (e.g., managing API rate limits) and the R econometric pipeline.

Critical Reflection and Responsible Usage: All generated code was manually written, reviewed, run locally, and validated against official documentation (e.g., Steam API, the *fixest* package in R). The underlying dataset architecture, econometric logic, and methodological choices remained entirely the responsibility of the author.

Instance of AI Usage:

Research Stage: Limitations

Purpose of AI Use and Tool Used: Gemini 1.5 Pro (Google) was used as an analytical sounding board to rigorously evaluate the methodological design and identify structural vulnerabilities, specifically regarding the constraints of using an aggregated FRED proxy, sample size limitations, and the Difference-in-Differences parallel trends assumption.

Critical Reflection and Responsible Usage: The AI functioned purely as a simulated peer-reviewer to pressure-test the empirical design. All highlighted limitations were critically evaluated by the author and integrated into the manuscript only if they accurately reflected the true boundaries and statistical constraints of the quantitative analysis.

D.3 Code Availability, README.md, and Replication Syntax

The full source code, data integration pipelines, and replication syntax for this study are hosted externally. The repository contains the custom automated data collection scripts, the local database initialization files, and the exact econometric models executed in R.

- **GitHub Repository:** https://github.com/totalmentecoder/dram-tracker
- **README.md:**

DRAM Price & AI Infrastructure Tracker

A live Python data pipeline correlating AI infrastructure demand with consumer DRAM pricing and PC game RAM requirements.

Built alongside my MSc thesis: *"Hardware Cannibalization and Software Strategy: AI Infrastructure Demand, DRAM Costs, and PC Game System Requirements"* – Tilburg University, 2025–2026.

The Thesis in One Chart

When NVIDIA's Data Center revenue overtook its Gaming revenue in 2022, semiconductor fabs pivoted toward High Bandwidth Memory (HBM) for AI GPUs. HBM and consumer DDR4/DDR5 compete for the **same silicon wafers** – so as AI demand surged, consumer DRAM supply tightened and prices broke their historical downward trend.

This project tracks that chain empirically:

```
NVIDIA Data Center Revenue ↑  →  HBM production priority ↑
                                 →  Consumer DRAM supply ↓
                                 →  DRAM prices ↑
                                 →  PC game RAM requirements plateau?
```

The research question: **Do game developers suppress hardware requirements and pay down technical debt when consumer RAM becomes expensive?**

What the Pipeline Does

Data Source	What it captures	API Details
FRED (Federal Reserve)	Semiconductor Producer Price Index (PCU334413334413) – manufacturing cost proxy for DRAM	Free, key required
NVIDIA IR	NVIDIA quarterly revenue: Data Center vs Gaming segment split loaded separately via quarterly CSV	Free
Steam Store API	Minimum & recommended RAM requirements per AAA title, release dates	Free, no auth, rate limited
SteamDB	Historical technical tags for AI upscaling adoption (DLSS, FSR, XESS)	Scraped, Manual
IGDB	Cross-platform title detection; console counterparts (category=0)	Free, Twitch auth
PCGamingWiki	Historical RAM requirements (2000–2022) for pre-shock baseline trend; verified manual overrides for titles	Free, Manual

All data is extracted, cleaned, and persisted to a local SQLite database (dram_tracker.db), which directly feeds the R econometric models.

Quickstart

```
# 1. Clone and install dependencies
git clone https://github.com/yourusername/dram-tracker.git
cd dram-tracker
pip install -r requirements.txt

# 2. Add your API keys (FRED free at https://fred.stlouisfed.org/docs/api/api_key.html)
#    Twitch credentials for IGDB at https://dev.twitch.tv/console
echo "FRED_API_KEY=your_key_here" > .env
echo "TWITCH_CLIENT_ID=your_id_here" >> .env
echo "TWITCH_CLIENT_SECRET=your_secret_here" >> .env

# 3. Run the pipeline
python pipeline.py
# → Opens interactive dashboard and saves dashboard.html
```

To skip re-fetching and just rebuild the dashboard from cached data:

```
from pipeline import init_db, build_dashboard
conn = init_db()
build_dashboard(conn).show()
```

Project Structure

```
dram-tracker/
├── pipeline.py                            # Main orchestration script — FRED, NVIDIA, Steam, SQLite, Plotly
├── igdb_collector.py                      # Platform detection — builds console control group
├── dlss_fsr_collector.py                  # AI upscaling integration tracker
├── games_list.py                          # Single source of truth for curated AAA game IDs
├── manual_loader.py                       # Applies manual CSV overrides to fix Steam API gaps
├── manual_overrides.csv                   # Verified RAM values from PCGamingWiki (13 titles)
├── build_its_dataset.py                   # Compiles the 72-row ITS sample
├── its_dataset_full.csv                   # OUTPUT: Full panel data for ITS model
├── build_did_dataset.py                   # Builds panel dataset for R DiD regression
├── did_dataset.csv                        # OUTPUT: Panel data for DiD model
├── nvidia_quarters.py                     # Parses NVIDIA quarterly segment revenue CSV
├── Formatted Data Revenue NVIDIA Quarterly.csv
├── pretrend/                              # Historical Wirth's Law baseline (2000-2022)
│   ├── pretrend.py
│   ├── pretrend_games.csv
│   └── Pretrend.png
├── analysis/                              # R scripts and regression outputs
│   ├── its_model.R                        # Primary Interrupted Time Series estimation (R/fixest)
│   ├── did_model.R                        # Supplementary DiD structural comparison
│   ├── h2_results.txt                     # Output: Logistic regression (Upscaling Adoption)
│   └── its_results.txt                    # Output: ITS Regression tables (H1)
├── dram_tracker.db                        # SQLite database (auto-created)
├── dashboard.html                         # Latest exported Plotly dashboard (auto-generated)
├── requirements.txt                       # Python dependencies
└── requirements_r_.txt                    # R dependencies (e.g., fixest)
```

Econometric Specifications & Estimation Strategy

The empirical architecture relies on two distinct specifications to map developer behavior across the Q3 2022 structural break.

1. Primary Model: Interrupted Time Series (ITS)

To evaluate the primary trajectory of software bloat, the pipeline estimates a segmented Ordinary Least Squares (OLS) model with heteroskedasticity-robust standard errors:

$$RAM\_it = \beta\_0 + \beta\_1 \cdot Time\_t + \beta\_2 \cdot Post\_t + \beta\_3 \cdot TimeSinceBreak\_t + \gamma \cdot Genre\_i + \epsilon\_it$$

- **β_3 (Primary Coefficient):** Captures the post-shock slope change to test for the suppression of hardware requirements.
- **Periodization:** Pre-shock baseline spans 2015–2021; the AI-intensive period spans 2023–2026. The year 2022 is explicitly excluded from primary estimation to prevent transition-year contamination around the mid-year shock.

2. Supplementary Model: Structural DiD Comparison

To capture platform governance dynamics, a descriptive Difference-in-Differences (DiD) framework is implemented. Because console observations are coded at a fixed 16 GB capacity dictated by static hardware cycles, this model operates as a descriptive measure of structural divergence rather than a strict causal identification check.

Three nested specifications are estimated within the R analysis pipeline (analysis/did_model.R):

1. **Specification 1 (Basic DiD):** Baseline model evaluating the interaction (PC_dummy × Post) with no additional covariates.
2. **Specification 2 (Genre Fixed Effects):** Introduces categorical controls to adjust for systematic baseline differences in memory demands across game categories.
3. **Specification 3 (Recommended RAM Covariate):** Incorporates recommended specs to control for the overall fidelity and scale of individual titles.

Key Findings (Final Thesis Outcomes)

- **Structural Break Confirmed:** A Chow test validates a statistically significant structural break in PC RAM requirements exactly at Q3 2022 ($p = 0.001$), coinciding precisely with the NVIDIA revenue segment crossover.
- **Wirth's Law Persists (H1):** Contrary to the hypothesis that hardware demands would suppress, PC RAM requirements accelerated. Mean minimum RAM jumped from 6.93 GB (pre-shock) to 11.40 GB (post-shock), representing a 64.5% increase.
- **Algorithmic Substitution (H2):** AI upscaling adoption exploded. Games released in the AI-intensive period are **20 times more likely** to integrate DLSS or FSR than pre-shock games (Odds Ratio = 19.7, $p < 0.001$).
- **Macroeconomic Proxy Masking (H3):** The predicted positive correlation between AI demand and consumer DRAM prices was not supported, as the broad FRED semiconductor PPI was dominated by long-run Moore's Law cost reductions (Spearman rho = -0.891, $p = 0.0004$). However, a localized upward anomaly in the index occurred precisely around the Q3 2022 break.
- **Substitution Mechanism (H4):** The direct test linking the price proxy to upscaling adoption was inconclusive. The sector-wide FRED PPI lacked sufficient variance (ranging only 2.6 index points) to validate the mechanism, highlighting the need for proprietary DDR4/DDR5 spot-price data in future research.
- **Structural Divergence (DiD):** The supplementary structural comparison confirmed that PC RAM requirements grew significantly more than the insulated, hardware-static console baseline, diverging by an additional +4.06 GB ($p = 0.0003$) following the supply shock.

Roadmap

[x] FRED DRAM price proxy pipeline

[x] NVIDIA revenue ingestion (yfinance)

[x] Steam Store API game requirements parser

[x] Manual override pipeline for Steam API gaps (13 PCGamingWiki-verified titles)

[x] IGDB integration for console counterparts (collector + schema complete)

[x] ITS & DiD dataset builders

[x] ITS & DiD regressions in R (fixest package) — completed and verified

[x] Historical pretrend chart (PCGamingWiki, 2000–2022)

[ ] Automated weekly refresh (GitHub Actions)

[ ] Power BI .pbix export integration

Data Notes & Limitations

- **FRED series PCU334413334413** is a Producer Price Index for the entire Semiconductor Manufacturing sector — not a pure DRAM spot price. True DRAM spot prices (DRAMeXchange/TrendForce) require a paid subscription; the PPI is used as a free, academically defensible proxy.
- **Steam API** rate limits require ~1.5s between requests. The pipeline includes a polite delay. Some delisted or free-to-play titles do not return data; these are corrected via manual_overrides.csv.
- **yfinance** provides consolidated NVIDIA revenue. Data Center vs. Gaming segment data is sourced from manually collected quarterly IR reports and loaded via nvidia_quarters.py.
- **Console RAM fixed at 16 GB** — PS5 and Xbox Series X both ship with 16 GB unified memory. This provides zero within-platform variation by design; all DiD identification comes from the PC treatment arm.
- The periodization (pre-AI: 2019–2022; AI-intensive: 2023–2026) reflects a conservative developer response lag. The underlying structural break in DRAM prices is identified at Q1 2022.
- **Pretrend analysis** uses a separate pretrend.db and does not modify dram_tracker.db.
- The periodization (pre-AI: 2015–2021; AI-intensive: 2023–2026) reflects a conservative developer response lag, with 2022 explicitly excluded from the primary ITS estimation to prevent transition-year contamination around the mid-year shock.

Tech Stack

Python 3.11 · SQLite · pandas · yfinance · fredapi · Plotly · requests · python-dotenv Analysis: R (fixest, tidyverse, ggplot2, modelsummary) Data: FRED API · Steam Store API · IGDB API · yfinance · PCGamingWiki

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